

Model Equivalence in Linear Non-Gaussian Causal Models under General Confounding

Yanchu Zhang

Wissenschaftliche Arbeit zur Erlangung des Grades

Master of Science

an der TUM School of Computation, Information and Technology
der Technischen Universität München

Prüfer:

Prof. Dr. Mathias Drton

Betreuer:

Prof. Dr. Mathias Drton

Eingereicht:

München, den 02 September 2026

Ich versichere hiermit, dass ich die von mir eingereichte Abschlussarbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Ort, Datum: München, 02.09.2026

Unterschrift: Yanhu Zhang

Acknowledgments

This thesis is a continuation of my student research project at the Chair of Mathematical Statistics, where I had the opportunity to be supervised by Daniele Tramontano and Prof. Mathias Drton. I would like to thank Daniele for introducing me to this topic and for his guidance and support during the early stages of my student research project. After Daniele completed his PhD, I had the opportunity to continue working on this topic for my master's thesis under the supervision of Prof. Mathias Drton. I am deeply grateful to Prof. Drton for his guidance, valuable feedback, and continuous support throughout this thesis and beyond. His encouragement and advice have been invaluable to my development as a researcher, and I am especially grateful for his generous support in helping me pursue the next steps of my academic career.

Abstract

This thesis investigates the central notion of model equivalence in causal discovery. If two models are equivalent, then no method based solely on observational data can, or should, distinguish between them. Consequently, characterizing model equivalence is fundamental as principled discovery methods rely on structural identifiability results.

Specifically, this thesis considers linear non-Gaussian causal models with general latent confounding, where latent effects are allowed to be nonlinear. In this important yet largely unexplored setting, model equivalence remains poorly understood, and no graphical characterization is currently known. This longstanding challenge has been a major obstacle to progress in causal discovery under general latent confounding. We address this gap by establishing, to the best of our knowledge, the first graphical characterization of model equivalence in this setting, and by developing algorithms for deciding equivalence in challenging cases.

Contents

Acknowledgment	i
Abstract	ii
Table of Contents	iii
1 Introduction	1
1.1 Motivation	1
1.2 Related Work	2
1.3 Thesis Organization	3
2 Problem Setup	3
2.1 Graphical Concepts	3
2.2 Model Definition	4
2.3 Model Equivalence	6
3 Main Results	7
3.1 Graphical Characterization	10
3.2 Towards a Simpler Characterization	18
3.3 Edge Removal	27
3.4 Edge Reversal	29
3.5 Constraints on the Confounding Structure	32
3.6 Computational Algebra for Cyclic Graphs	37
3.7 Solving via Satisfiability Modulo Theories	43
4 Conclusion	45
4.1 Limitations and Future Work	46
A Code Repository	53

1 Introduction

1.1 Motivation

Graphical models are widely used to represent causal relationships among observed and, potentially, unobserved variables [Spirtes, 1994, Lauritzen, 1996, Pearl, 2013, Maathuis et al., 2019]. In practice, the underlying causal graph is often unknown. Causal discovery addresses the fundamental problem of inferring the graph from data [Drton and Maathuis, 2017, Glymour et al., 2019]. Central to causal discovery is the notion of model equivalence: if two models are equivalent, then no method based solely on observational data can, or should, distinguish between them. Consequently, characterizing model equivalence is essential for causal discovery as principled methods rely on structural identifiability results [Adams et al., 2021, Dai et al., 2026]. Well-known examples include the PC algorithm being based on d-separation for Markov equivalence classes [Spirtes et al., 2001] and FCI’s guarantee with respect to partial ancestral graphs (PAGs) [Richardson and Spirtes, 2002, Zhang, 2008].

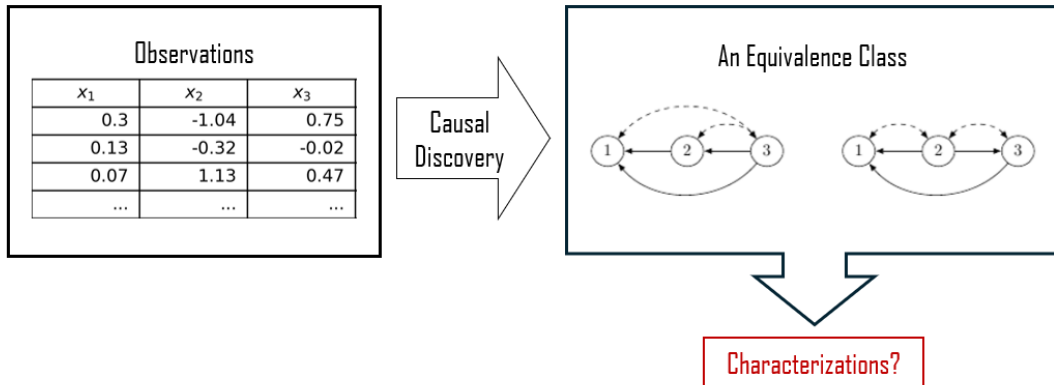


Figure 1: An overview of the relationships between causal discovery, equivalence classes, and characterizations.

A major challenge in causal discovery is understanding models with latent confounding. A latent confounder is an unobserved variable that induces dependencies among the observed variables. In the presence of latent confounding, the noise terms are generally no longer independent, and a unique causal graph cannot, in general, be identified from observational data. Instead, only an equivalence class of causal graphs, all of which are compatible with the same set of observational distributions, can be recovered. Therefore, it is essential to understand how these equivalent causal structures can be characterized (Figure 1).

In this work, we consider linear non-Gaussian models under general latent confounding, where latent effects are allowed to be nonlinear [Wang and Drton, 2023, Tramontano et al., 2026]. In this important yet largely unexplored setting, model equivalence remains poorly understood, and no graphical characterization of equivalence is currently known. This challenge has remained a major obstacle to progress in causal discovery under general latent confounding. We address this gap by establishing, to the best of our knowledge, the first graphical criterion for model equivalence under observed data in this minimally restrictive linear setting, and by developing algorithms for deciding equivalence in challenging cases.

Table 1: Comparison of related work on the equivalence of models under non-Gaussian noise and observational distribution.

Reference	Linear Confounding	Nonlinear Confounding	Structural Assumption	Graphical Object
Shimizu et al. [2006]	✗	✗	acyclic	directed acyclic graph (DAG)
Liu et al. [2021]	✓	✓	acyclic, bow-free	mixed graph with multi-directed edges
Wang and Drton [2023], van Ommen [2024]	✓	✓	acyclic, bow-free	bow-free acyclic path diagram (BAP)
Drton et al. [2025a]	✗	✗	allow disjoint cycles	directed graph (DG)
Dai et al. [2026]	✓	✗	allow cycles	DG
Graphical characterization (Theorem 3.7)	✓	✓	acyclic	acyclic directed mixed graph (ADMG)
Edge removal (Theorem 3.23) and edge reversal (Theorem 3.24)	✓	✓	acyclic	ADMG
Confounding structure (Theorem 3.28)	✓	✓	allow cycles	directed mixed graph (DMG)
Computer algebra (Algorithm 2)	✓	✓	allow cycles	DMG

1.2 Related Work

There is a wide range of work characterizing equivalence under various settings. For Markov equivalence alone, this has been studied in fully observed settings [Frydenberg, 1990, Verma and Pearl, 1990, Spirtes and Glymour, 1991, Andersson et al., 1997, Roverato and Studeny, 2006] and in settings with latent variables [Spirtes and Verma, 1992, Spirtes and Richardson, 1996, Ali et al., 2005, Hu and Evans, 2020]. Other works focus on cyclic models [Claassen and Mooij, 2023], discrete settings [Pearl and Dechter, 1996], or general nonlinear settings [Mooij and Claassen, 2020]. Moreover, distributional equivalence in fully observed linear Gaussian cyclic models has been studied in Ghassami et al. [2020], Drton et al. [2025b]; nonparametric equivalence with latent variables has been characterized in Evans [2018], Richardson et al. [2023]; and Markov equivalence under interventions or domain knowledge has been studied in Tian and Pearl [2001], Rothenhäusler et al. [2015].

Among the various settings studied, linear non-Gaussian models have attracted considerable attention due to their favorable identifiability properties. Under the assumption of no hidden confounding, linear non-Gaussian acyclic models (LiNGAMs) are structurally identifiable and can be uniquely recovered from the observational distribution [Shimizu et al., 2006, Shimizu, 2022]. Attention has subsequently shifted to settings with latent confounding. In particular, the effects of latent variables may be modeled either linearly or nonlinearly, with most existing work focusing on linear latent confounding. For linear non-Gaussian models, advances in overcomplete independent component analysis (OICA) have enabled finer-grained structure learning beyond what can be inferred from conditional independence constraints alone [Salehkaleybar et al., 2020,

Maeda and Shimizu, 2020, Liu et al., 2021, Améndola et al., 2023, Tramontano et al., 2024, Schkoda et al., 2024, Yang et al., 2024, Drton et al., 2025a]. The extensively studied latent variable linear non-Gaussian model (lvLiNG) falls within this category, but its equivalence results have the limitation of assuming an explicit linear latent structure a priori [Dai et al., 2026] (see Figure 20 for a comparison).

Significant challenges remain in the presence of nonlinear hidden confounding. In this setting, latent confounders can be naturally represented by bidirected edges, without requiring explicit relationships among the latent variables. The setting considered in this thesis allows for general confounding, where the confounding effects can be both linear and nonlinear. This setting is largely aligned with that of Tramontano et al. [2026], which investigates the identifiability of direct causal effects. Existing results on model equivalence in the presence of general confounding rely on strong structural assumptions, such as bow-freeness of the underlying graph [Wang and Drton, 2023, van Ommen, 2024]. A general characterization of model equivalence without imposing such assumptions has remained elusive. Addressing this fundamental open problem is the primary objective of the present work.

For a comparison of related work on equivalence in linear non-Gaussian models across different settings, see Table 1.

1.3 Thesis Organization

The thesis is organized as follows. We begin by introducing the basic graphical concepts in Section 2.1, with particular emphasis on the terminology associated with (acyclic) directed mixed graphs. We then define the general confounded linear non-Gaussian model in Section 2.2. We complete the problem setup by introducing two notions of equivalence in Section 2.3.

We then present our main results. We first establish a characterization of equivalence (Theorem 3.1) based on the solvability of a system of quadratic equations. This result provides the foundation for the subsequent developments. In Section 3.1, we derive the first graphical characterization of equivalence. We then further simplify the characterization, implement the resulting algorithm, and evaluate its performance in Section 3.2. For local criteria on equivalence, we study edge removal in Section 3.3 and edge reversal in Section 3.4. In Section 3.5, we study constraints on the confounding structure using cumulants. We demonstrate a computer algebra based approach in Section 3.6, which accommodates cyclic graphs. Additionally, we discuss an alternative constraint programming based approach, which demonstrates superior performance compared to the previous methods in Section 3.7. Finally, in Section 4, we evaluate our contributions and limitations, and conclude by discussing directions for future research.

2 Problem Setup

2.1 Graphical Concepts

The central graphical objects we study are (acyclic) directed mixed graphs, abbreviated as (A)DMGs. A directed mixed graph is a triple $G = (V, E_{\rightarrow}, E_{\leftrightarrow})$, where V denotes the set of vertices (or nodes) and $E_{\rightarrow}, E_{\leftrightarrow} \subseteq V \times V \setminus \{(v, v) : v \in V\}$ denote the set of directed and bidirected edges, respectively. The bidirected edge set $E_{\leftrightarrow}$ needs to be symmetric. In this thesis, we will often write $p := |V|$ for the number of vertices. Sometimes we want to emphasize only a specific part of a DMG G ; namely, we write G_D for the directed part $(V, E_{\rightarrow})$ and we write G_B for the bidirected part $(V, E_{\leftrightarrow})$. If it is clear from the context, we will for simplicity just write $u \rightarrow v \in G$ (or $u \rightarrow v \in G_D$) and $u \leftrightarrow v \in G$ (or $u \leftrightarrow v \in G_B$) for $u \rightarrow v \in E_{\rightarrow}$ and $u \leftrightarrow v \in E_{\leftrightarrow}$ with $E_{\rightarrow}, E_{\leftrightarrow} \in G$.

Let $u, v \in V$ be two vertices in G . A directed path from u to v is a sequence of nodes $\pi = (v_1, \dots, v_k)$ with $v_1 = u$, $v_k = v$ such that $v_i \rightarrow v_{i+1} \in E_{\rightarrow}$ for all $i \in \{1, \dots, k-1\}$. A bidirected path is defined analogously by considering $v_i \leftrightarrow v_{i+1} \in E_{\leftrightarrow}$. We say that a path is **trivial** if $\pi = u$ contains only a single node and no edge. A directed circle is a non-trivial directed path from a node to itself. If a graph contains no directed circle, then it is **acyclic**. This special class of DMGs are referred to as **acyclic directed mixed graphs** (ADMGs).

Let $\mathcal{P}(u, v)$ be the set of all directed paths from u to v . The following genealogical relations of vertices are commonly used:

- (parents) $\text{pa}(v) := \{u \in V : u \rightarrow v \in G\}$,
- (children) $\text{ch}(v) := \{u \in V : v \rightarrow u \in G\}$,
- (ancestors) $\text{an}(v) := \{u \in V : \mathcal{P}(u, v) \neq \emptyset \in G\}$,
- (descendants) $\text{de}(v) := \{u \in V : \mathcal{P}(v, u) \neq \emptyset \in G\}$,
- (siblings) $\text{sib}(v) := \{u \in V : v \leftrightarrow u \in G\}$,
- $\text{Sib}(v) := \text{sib}(v) \cup \{v\}$.

By definition, we have $v \in \text{an}(v)$ and $v \in \text{de}(v)$, via the trivial path. For a subset $U \subseteq V$, we define $\text{pa}(U) = \cup_{u \in U} \text{pa}(u)$ and make analogous convention for the other relations.

2.2 Model Definition

Let $G = (V, E_{\rightarrow}, E_{\leftrightarrow})$ be a DMG with $p = |V|$. Let $(X_v)_{v \in V}$ be a finite collection of observed random variables. A linear model given by G postulates that each variable is a linear function of its own parents and a noise term:

$$X_v = \sum_{u \in \text{pa}(v)} \lambda_{uv} X_u + \varepsilon_v, \quad v \in V. \quad (2.1)$$

The direct effect $\lambda_{uv} \in \mathbb{R}$ is set to be zero if there is no directed edge from u to v . We summarize them in a matrix $\Lambda = (\lambda_{uv}) \in \mathbb{R}^{p \times p}$ and define the set of directed effect matrices associated with G formally as

$$\mathbb{R}^{G_D} := \{\Lambda \in \mathbb{R}^{p \times p} : \Lambda_{uv} = 0 \text{ if } u \rightarrow v \notin G_D\}. \quad (2.2)$$

Note that λ_{uv} can still be zero under presence of $u \rightarrow v$.

The next step is to make assumptions on the non-Gaussian noise $\varepsilon \in \mathbb{R}^p$. Assuming non-Gaussian noise is central as it yields favorable identifiability properties. It has also been argued that the non-Gaussianity assumption is primarily an empirical assumption in the presence of heteroscedasticity [Hyvärinen et al., 2010]. A subset $C \subseteq V$ is **connected** in G_B if for every pair $u, v \in C$, there is a bidirected path in G_B that consists only of vertices in C . We now introduce the first assumption on the noise ε : **the connected set Markov property** [Drton and Richardson, 2008, Richardson, 2003].

Definition 2.1 (Connected set Markov property). *A random vector ε satisfies the connected set Markov property with respect to G_B if for all non-empty subsets $C \subseteq V$, we have $\varepsilon_C \perp\!\!\!\perp \varepsilon_{V \setminus \text{Sib}(C)}$ whenever C is connected in G_B .*

This is a natural generalization of the pairwise version, in which that each bidirected edge $u \leftrightarrow v$ indicates possibly (nonlinear) dependence between ε_u and ε_v .

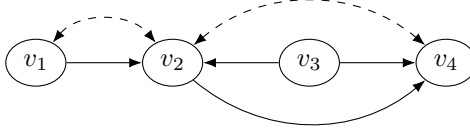


Figure 2: An example of an ADMG with four nodes.

Example 2.1. Under the model given by the ADMG in Figure 2, we have

$$X_1 = \varepsilon_1, \quad X_2 = \lambda_{12}X_1 + \lambda_{32}X_3 + \varepsilon_2, \quad (2.3)$$

$$X_3 = \varepsilon_3, \quad X_4 = \lambda_{34}X_3 + \lambda_{24}X_2 + \varepsilon_4. \quad (2.4)$$

The graph allows for the dependences $\varepsilon_1 \not\perp \varepsilon_2$ and $\varepsilon_2 \not\perp \varepsilon_4$.

The next nonparametric assumption imposed on the noise term is natural, yielding benign statistical properties while remaining minimally restrictive.

Definition 2.2 (Genericity of ε). *A vector ε is generic if for arbitrary $a, b \in \mathbb{R}^p$, it holds that*

$$a^T \varepsilon \perp b^T \varepsilon \implies a_u \cdot b_v = 0 \quad \text{for all } u = v \text{ and } u \leftrightarrow v \in E_{\leftrightarrow}. \quad (2.5)$$

This genericity assumption is natural in view of the Darmois-Skitovich theorem [Darmois, 1953, Skitovich, 1954], which in our setting amounts to the fact that for $E_{\leftrightarrow} = \emptyset$, Equation (2.5) is true, provided the random variable ε is non-Gaussian (strictly speaking, the equation above holds if at most one of the random variables ε_v is Gaussian). We state the well-known theorem below:

Theorem 2.1 (Darmois-Skitovich). *Let $\varepsilon = (\varepsilon_1, \dots, \varepsilon_p)^T$ be a random variable with jointly independent components $\varepsilon_1, \dots, \varepsilon_p$. Let $a, b \in \mathbb{R}^p$ be two constant vectors with all entries nonzero. Then it holds that*

$$a^T \varepsilon \perp b^T \varepsilon \implies \varepsilon_1, \dots, \varepsilon_p \text{ are all Gaussian.} \quad (2.6)$$

The idea behind the generalization to $E_{\leftrightarrow} \neq \emptyset$ is to prohibit the possibility of linearly disentangling a general dependence between two errors ε_v and ε_w . It can be shown that Equation (2.5) is satisfied by general distributions. In particular, when considering distributions with finite moments up to a suitable large order or linear latent factor models, the genericity assumption is indeed almost always satisfied; we refer to Tramontano et al. [2026] for further details.

Definition 2.3. *Given a DMG G , the set $\mathcal{M}(G_B)$ consists of non-Gaussian noise vectors $\varepsilon \in \mathbb{R}^p$ that satisfy both the connected set Markov property with respect to G_B and the genericity assumption.*

Equations such as those in (2.1) can eventually be written as structural equations of the form $X = \Lambda^T X + \varepsilon$. If Λ arises from an ADMG, then $I - \Lambda$ is always invertible; we can solve the system of equations and write $X = (I - \Lambda)^{-T} \varepsilon$. If Λ arises from a DMG G , then $I - \Lambda$ can be singular for some specific parameter choice, which forms a null set of the parameter space. In the following, we will always assume that $I - \Lambda$ is invertible by taking Λ from the regulated space $\mathbb{R}_{\text{reg}}^{G_D} := \{\Lambda \in \mathbb{R}^{G_D} : \det(I - \Lambda) \neq 0\}$. And for simplicity, we will from now on write $\mathbb{R}^{G_D}$ for $\mathbb{R}_{\text{reg}}^{G_D}$ whenever it is clear from context. Let $B_\Lambda := (I - \Lambda)^{-T}$, and we obtain $X = B_\Lambda \varepsilon$. We remark that the matrix B_Λ has the following combinatorial property if Λ is acyclic. Its entry b_{uv} is determined by all the directed paths from v to u :

Lemma 2.2 (Sullivant et al. [2010]). *Let G be an ADMG, then*

$$(B_\Lambda)_{uv} = \sum_{\pi \in \mathcal{P}(v,u)} \prod_{w_i \rightarrow w_j \in \pi} \lambda_{ij}, \quad \forall v \in \text{an}(u) \setminus u. \quad (2.7)$$

Moreover, $(B_\Lambda)_{uu} = 1$ and $(B_\Lambda)_{uv} = 0$ if $v \notin \text{an}(u)$.

For a cyclic DMG G , the path matrix interpretation of B_Λ becomes a formal power series. We are now finally prepared to define our model:

Definition 2.4 (LiNG). *Let G be a DMG. The general confounded linear non-Gaussian model (LiNG) associated with G , denoted by $\mathcal{M}(G)$, is given by the following parametrization:*

$$\mathcal{M}(G) = \{X \in \mathbb{R}^p : X \stackrel{d}{=} (I - \Lambda)^{-T} \varepsilon, (\Lambda, \varepsilon) \in \mathbb{R}^{G_D} \times \mathcal{M}(G_B)\}. \quad (2.8)$$

2.3 Model Equivalence

In the presence of hidden confounding, given observational data, one can at best infer an equivalence class, that is, a collection of causal graphs associated with the same set of probability distributions. Understanding the equivalence of causal models is therefore fundamental to the development of any principled causal discovery method. We now formally introduce this concept.

Let G and $\tilde{G}$ be two arbitrary DMGs with the same vertex set V . Then $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ if and only if for all $X \in \mathcal{M}(G)$, there exist some $(\tilde{\Lambda}, \tilde{\varepsilon}) \in \mathbb{R}^{\tilde{G}_D} \times \mathcal{M}(\tilde{G}_B)$ such that $X \stackrel{d}{=} (I - \tilde{\Lambda})^{-T} \tilde{\varepsilon}$, where $\stackrel{d}{=}$ denotes equal in distribution. We write $\mathcal{M}(G) = \mathcal{M}(\tilde{G})$ if $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ and $\mathcal{M}(\tilde{G}) \subseteq \mathcal{M}(G)$. We now obtain the strict version of model equivalence.

Definition 2.5 (Strictly model equivalence). *Two DMGs G and $\tilde{G}$ are strictly model equivalent if $\mathcal{M}(G) = \mathcal{M}(\tilde{G})$.*

This first version of model equivalence is natural in view of the definition, but sometimes overly strict, as it requires solvability of $(\tilde{\Lambda}, \tilde{\varepsilon}) \in \mathbb{R}^{\tilde{G}_D} \times \mathcal{M}(\tilde{G}_B)$ for all $X \in \mathcal{M}(G)$, and vice versa. Generally speaking, we want two DMGs to be model equivalent if they are “almost” associated with the same set of distributions, meaning that we allow some measure-zero exceptions; in that case, the two models are still statistically indistinguishable from observational data. This motivates our next version of model equivalence:

Definition 2.6 (Model equivalence (up to closure)). *For two DMGs G and $\tilde{G}$, we say that $\mathcal{M}(G)$ is contained in the closure of $\mathcal{M}(\tilde{G})$, denoted by $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$, if there is a null set $N \subseteq \mathbb{R}^{G_D}$ such that every joint distribution $X = (I - \Lambda)^{-T} \varepsilon$ with $(\Lambda, \varepsilon) \in (\mathbb{R}^{G_D} \setminus N) \times \mathcal{M}(G_B)$ is in $\mathcal{M}(\tilde{G})$. We say that G and $\tilde{G}$ are model equivalent up to closure (or simply model equivalent), if $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$ and $\mathcal{M}(\tilde{G}) \subseteq \overline{\mathcal{M}(G)}$; we denote it by $\overline{\mathcal{M}(G)} = \overline{\mathcal{M}(\tilde{G})}$.*

The central topic we study in this work is model equivalence, and we introduced two versions of it. Note that the difference is, however, statistically minimal: the strict one requires overall solvability of the parameters, whereas the relaxed one allows an exceptional Lebesgue null set, corresponding to generic solvability of the parameters; this point will be made more clear after introducing our first result. Another motivation of considering the closure is to find graphical characterizations of equivalent models. The common tools, including the rank condition which bridges linear algebra and graph theory, only captures equivalence up to closure. Note that excluding null sets of a finite-dimensional space is a standard approach in the related literature, not only when considering equivalence of models [Adams et al., 2021, van Ommen, 2024, Dai et al.,

2026], but also when considering identifiability of parameters [Drton, 2018, Tramontano et al., 2026]. In the sequel, we make the convention that model equivalence (or simply “equivalence”) should be understood as model equivalence up to closure.

3 Main Results

We now seek a tractable characterization of model equivalence. Since two models are equivalent if and only if they are mutually inclusive, it suffices to characterize model inclusion. A parameter choice is called **generic** if it lies outside a Lebesgue null set in the parameter space. Our first result is the following theorem which checks model inclusion by converting a general parametrization problem into a system of quadratic equations.

Theorem 3.1. *Let G and $\tilde{G}$ be two DMGs with the same vertex set V . Then*

- $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ if and only if for all $\Lambda \in \mathbb{R}^{G_D}$, the system $\mathcal{L}$ below is solvable in $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$.
- $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$ if and only if for any generic $\Lambda \in \mathbb{R}^{G_D}$, the system $\mathcal{L}$ below is solvable in $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$.

Here we define

$$\mathcal{L} := \{A_{uw_1} \cdot A_{vw_2} = 0 : (u, v) \in \mathcal{J}, (w_1, w_2) \in \mathcal{K}\}, \quad (3.1)$$

$$A := (I - \tilde{\Lambda})^T (I - \Lambda)^{-T}, \quad (3.2)$$

$$\mathcal{J} := \{(u, v) : u \neq v \text{ and } u \leftrightarrow v \notin \tilde{E}_{\leftrightarrow}\} \text{ and} \quad (3.3)$$

$$\mathcal{K} := \{(w_1, w_2) : w_1 = w_2 \text{ or } w_1 \leftrightarrow w_2 \in E_{\leftrightarrow}\}. \quad (3.4)$$

We remark that A_{ij} is an affine linear function of $\tilde{\lambda}_{\bullet, i}$, where $\tilde{\lambda}_{\bullet, i}$ denotes the collection $\{\lambda_{ji} : j \in V \setminus i\}$. Therefore, the product $A_{uw_1} \cdot A_{vw_2}$ is at most quadratic in $\tilde{\Lambda}$. We see that a quadratic term involving a single variable, such as $\tilde{\lambda}_{ki}^2$, cannot appear in the product, since this would imply $u = v$, contradicting $(u, v) \in \mathcal{J}$. Nevertheless, quadratic terms involving two distinct variables do appear in the equations. Moreover, $\mathcal{L}$ can be viewed as a particular instance of an edge ideal [cf. Herzog and Hibi, 2011, Section 9.1].

Proof. We start with the claim on the strict version of model equivalence. By definition, $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ if and only if for every $X = (I - \Lambda)^{-T} \varepsilon \in \mathcal{M}(G)$, we have $X \stackrel{d}{=} (I - \tilde{\Lambda})^{-T} \tilde{\varepsilon}$ for some $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$ and $\tilde{\varepsilon} \in \mathcal{M}(\tilde{G}_B)$. Note that we assume that both $I - \Lambda$ and $I - \tilde{\Lambda}$ are invertible. Then this implies

$$\tilde{\varepsilon} \stackrel{d}{=} (I - \tilde{\Lambda})^T (I - \Lambda)^{-T} \varepsilon = \underbrace{(I - \tilde{\Lambda})^T B_{\Lambda}}_A \varepsilon. \quad (3.5)$$

In particular, Equation (3.5) has to hold for $\varepsilon \in \mathcal{M}(G_B)$ satisfying Equation (2.5). Choose $a, b \in \mathbb{R}^p$ such that a_u and b_v are both nonzero only for $u \leftrightarrow v \notin \tilde{G}$ and $u \neq v$. Then $a^T \tilde{\varepsilon} \perp b^T \tilde{\varepsilon}$ by model definition. Since $\tilde{\varepsilon} \stackrel{d}{=} A\varepsilon$, we also have $a^T (A\varepsilon) \perp b^T (A\varepsilon)$ for this choice of a, b . Applying the genericity assumption (2.5), we obtain that the product $(a^T A)_{w_1} \cdot (b^T A)_{w_2} = 0$ for $w_1 = w_2$ or $w_1 \leftrightarrow w_2 \in G$, and conclude that the condition is necessary for the model inclusion.

In order to prove its sufficiency, we must demonstrate that $\tilde{\varepsilon}$ satisfies the connected set Markov property with respect to $\tilde{G}$, that is $\tilde{\varepsilon}_C \perp \tilde{\varepsilon}_{V \setminus \text{Sib}(C)}$ for every $C \subseteq V$ connected in $\tilde{G}_B$. Equivalently, one can show for such C that $(A\varepsilon)_C \perp (A\varepsilon)_{V \setminus \text{Sib}(C)}$. This amounts to checking that the remaining corresponding nonzero coefficients do not create dependencies, namely, for

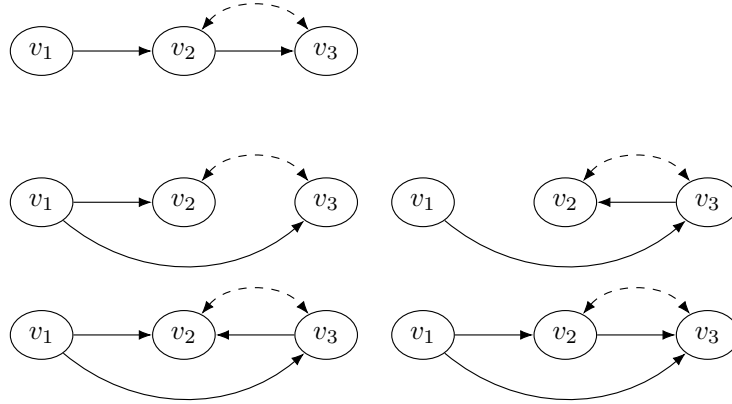


Figure 3: The equivalence class of the IV graph. Upper left: G_1 (IV Graph), middle left: G_2 , middle right: G_3 , bottom left: G_4 , bottom right: G_5 .

any connected $C \in V$ in $\tilde{G}_B$, we must have $\sum_{j \in V} A_{uj} \varepsilon_j \perp \sum_{k \in V} A_{vk} \varepsilon_k$ with $u \in C$ and $v \in V \setminus \text{Sib}_{\tilde{G}}(C)$. This independence is true if at least one of A_{uw_1} and A_{vw_2} is zero for each pair of $w_1 = w_2$ or $w_1 \leftrightarrow w_2 \in G$, by connected set Markov property with respect to ε in G .

The statement on model equivalence (up to closure) follows directly by considering the definition of generic solvability. The system $\mathcal{L}$ is solvable in $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$ for any generic $\Lambda \in \mathbb{R}^{G_D}$ if and only if there is an exceptional null set $\mathbf{N} \subseteq \mathbb{R}^{G_D}$ so that $\mathcal{L}$ is solvable in $\tilde{\Lambda}$ for all $\mathbb{R}^{G_D} \setminus \mathbf{N}$. Following the same proof, parametrization from a model $\mathcal{M}(G)$ into model $\mathcal{M}(\tilde{G})$ is possible except on such null set $\mathbf{N}$. $\square$

We note that the system $\mathcal{L}$ has redundant equations, because if $A_{uw_1} A_{vw_2} = 0$ is in $\mathcal{L}$, then $A_{vw_2} A_{uw_1} = 0$ is also in $\mathcal{L}$. Without loss of generality, we will restrict ourselves from $\mathcal{J}$ to $\mathcal{J}^<$ by only allowing pairs (u, v) with $u < v$, and write $\mathcal{J}$ for $\mathcal{J}^<$. The next lemma investigates the entries of the matrix A .

Lemma 3.2. *Let A be defined as in Equation (3.2). Then*

$$A_{ij} = b_{ij} - \sum_{k \in V \setminus i} \tilde{\lambda}_{ki} b_{kj}, \quad (3.6)$$

where $b_{ij} = (I - \Lambda)_{ij}^{-T}$.

Proof. For a matrix B , we write B_i to denote the i -th column of B , then $A_{ij} = \langle (I - \tilde{\Lambda})_i, (B_\Lambda)_j \rangle$. The result follows directly by noticing that $((I - \tilde{\Lambda})^T)_{ii} = 1$ and $((I - \tilde{\Lambda})^T)_{ik} = -\tilde{\lambda}_{ki}$ for $i \neq k$. $\square$

Example 3.1. Consider the simplest possible case, where G and $\tilde{G}$ are DAGs on two nodes, $V = \{v_1, v_2\}$, with G having the single edge $v_1 \rightarrow v_2$ and $\tilde{G}$ having only the reversed edge $v_2 \rightarrow v_1$. To check whether $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$, we first construct the matrix

$$A = (I - \tilde{\Lambda})^T (I - \Lambda)^{-T} = \begin{pmatrix} 1 - \tilde{\lambda}_{21} \lambda_{12} & -\tilde{\lambda}_{21} \\ \lambda_{12} & 1 \end{pmatrix}. \quad (3.7)$$

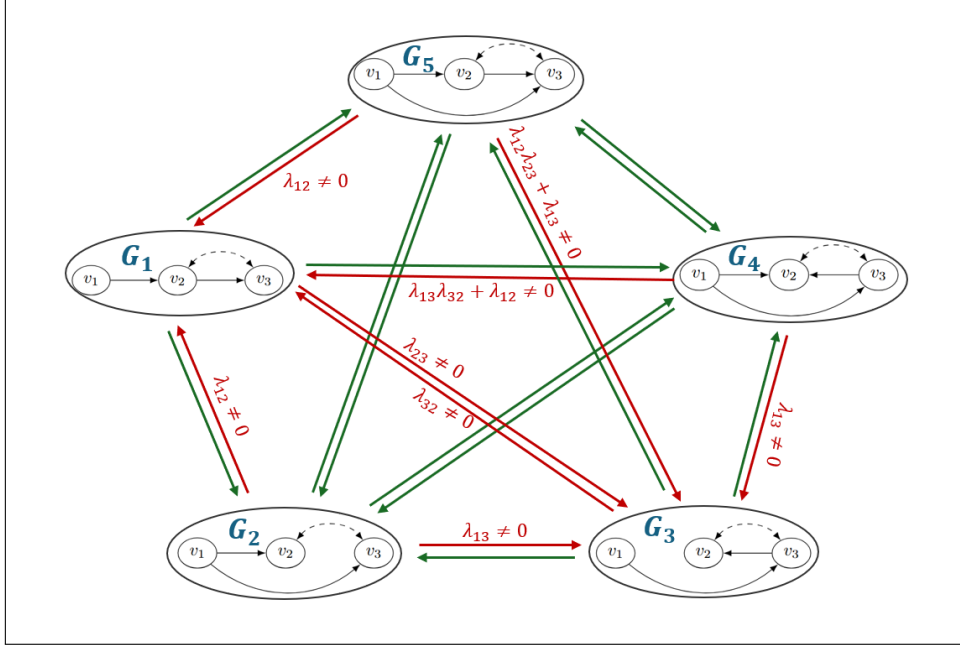


Figure 4: IV equivalence class in detail. A green arrow from G_i to G_j indicates that $\mathcal{M}(G_i) \subseteq \mathcal{M}(G_j)$. A red arrow from G_i to G_j indicates $\mathcal{M}(G_i) \subseteq \overline{\mathcal{M}(G_j)}$; the corresponding generic parameter choice is indicated above the arrow.

Since $v_1 \leftrightarrow v_2 \notin \tilde{G}$, we need to check if the equation system

$$L = \{A_{11} \cdot A_{21} = 0, A_{12} \cdot A_{22} = 0\} \quad (3.8)$$

$$= \{(1 - \tilde{\lambda}_{21}\lambda_{12})\lambda_{12} = 0, -\tilde{\lambda}_{21} = 0\} \quad (3.9)$$

has a solution in $\tilde{\lambda}_{21} \in \mathbb{R}$ for all $\lambda_{12} \in \mathbb{R}$. Evidently, L has solution $\tilde{\lambda}_{21} = 0$ when $\lambda_{12} = 0$, but L has no solution when $\lambda_{12} \neq 0$. We may deduce that $\mathcal{M}(G) \not\subseteq \overline{\mathcal{M}(\tilde{G})}$, and by symmetry, $\mathcal{M}(G) \not\subseteq \overline{\mathcal{M}(\tilde{G})}$. It is clear that this also implies $\mathcal{M}(G) \not\subseteq \mathcal{M}(\tilde{G})$ and $\mathcal{M}(G) \not\subseteq \overline{\mathcal{M}(\tilde{G})}$.

For a less trivial illustration of Theorem 3.1, we consider graphs related to the **instrumental variable (IV)** graph (Figure 3).

Example 3.2. Consider the graphs from Figure 3. Using Theorem 3.1, we may show that $\mathcal{M}(G_2) = \mathcal{M}(G_4) = \mathcal{M}(G_5) \neq \{\mathcal{M}(G_1) \text{ or } \mathcal{M}(G_3)\}$, and $\overline{\mathcal{M}(G_1)} = \overline{\mathcal{M}(G_2)} = \overline{\mathcal{M}(G_3)} = \overline{\mathcal{M}(G_4)} = \overline{\mathcal{M}(G_5)}$ (see Figure 4). To exemplify one of the computations, we show that $\mathcal{M}(G_3) \neq \mathcal{M}(G_2)$ but $\overline{\mathcal{M}(G_3)} = \overline{\mathcal{M}(G_2)}$. Consider $G := G_3$ and $\tilde{G} := G_2$, starting with checking whether $\mathcal{M}(G_3) \subseteq \mathcal{M}(G_2)$, we calculate

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \lambda_{13}\lambda_{32} - \tilde{\lambda}_{12} & 1 & \lambda_{32} \\ -\tilde{\lambda}_{13} + \lambda_{13} & 0 & 1 \end{pmatrix}. \quad (3.10)$$

Since $A_{12}, A_{13} = 0$, the system of equations simplifies to

$$L = \{A_{11} \cdot A_{21} = 0, A_{11}A_{31} = 0\} \quad (3.11)$$

$$= \{\lambda_{13}\lambda_{32} - \tilde{\lambda}_{12} = 0, -\tilde{\lambda}_{13} + \lambda_{13} = 0\}. \quad (3.12)$$

We see that L is solved by $\tilde{\lambda}_{12} = \lambda_{13}\lambda_{32}$ and $\tilde{\lambda}_{13} = \lambda_{13}$ for all $\Lambda \in \mathbb{R}^{G_D}$. We conclude $\mathcal{M}(G_3) \subseteq \mathcal{M}(G_2)$ and therefore also $\mathcal{M}(G_3) \subseteq \overline{\mathcal{M}(G_2)}$.

For the other direction, consider $G := G_2$ and $\tilde{G} := G_3$. To check whether $\mathcal{M}(G_2) \subseteq \mathcal{M}(G_3)$ now, we calculate

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \lambda_{12} - \tilde{\lambda}_{32}\lambda_{13} & 1 & -\tilde{\lambda}_{32} \\ -\tilde{\lambda}_{13} + \lambda_{13} & 0 & 1 \end{pmatrix}. \quad (3.13)$$

The system of equations simplifies to

$$L = \{A_{11} \cdot A_{21} = 0, A_{11}A_{31} = 0\} \quad (3.14)$$

$$= \{\lambda_{12} - \tilde{\lambda}_{32}\lambda_{13} = 0, -\tilde{\lambda}_{13} + \lambda_{13} = 0\}. \quad (3.15)$$

If $\lambda_{13} \neq 0$, L has solution $\tilde{\lambda}_{13} = \lambda_{13}$ and $\tilde{\lambda}_{32} = \frac{\lambda_{12}}{\lambda_{13}}$. Therefore, $\mathcal{M}(G_2) \subseteq \overline{\mathcal{M}(G_3)}$. However, if $\lambda_{13} = 0$ and $\lambda_{12} \neq 0$, then L has no solution. Due to these exceptional cases, $\mathcal{M}(G_2) \not\subseteq \mathcal{M}(G_3)$.

3.1 Graphical Characterization

Since our statistical models are entirely determined by their associated graphs, it is natural to investigate when two models are graphically equivalent. To this end, we first establish that model equivalence up to closure admits a characterization in terms of a system of linear equations involving a special selection operation, which we call a “split”. This selection operation enables the original equations to be transformed directly into a linear system. This characterization will subsequently lead to graphical results.

To formalize this, we first define the reindexed Cartesian product $\mathcal{J} \otimes \mathcal{K} := \{(u, w_1, v, w_2) \in V^4 : (u, v) \in \mathcal{J}^<, (w_1, w_2) \in \mathcal{K}\}$ with $\mathcal{J}, \mathcal{K}$ defined as in Equations (3.3) and (3.4). For any $\mathbf{S} \subseteq \mathcal{J} \otimes \mathcal{K}$, we let $\mathbf{S}[1]$ be the image of applying the projection $(u, w_1, v, w_2) \mapsto (u, w_1)$ to each of the element in $\mathbf{S}$. Similarly, we let $\mathbf{S}[2]$ be the image of applying $(u, w_1, v, w_2) \mapsto (v, w_2)$ to $\mathbf{S}$.

Definition 3.1 (Confounding split). *Let G and $\tilde{G}$ be two DMGs with the same vertex set V . We call $H_S : \mathbf{S} \mapsto \mathbf{S}[1] \cup (\mathcal{J} \otimes \mathcal{K} \setminus \mathbf{S})[2]$ a split function with $\mathbf{S} \subseteq \mathcal{J} \otimes \mathcal{K}$. Given $G, \tilde{G}$, we say $\mathcal{H} \subseteq V^2$ is a confounding split if this can be obtained by a split function with any $\mathbf{S} \subseteq \mathcal{J} \otimes \mathcal{K}$.*

Note that $|\mathcal{J} \otimes \mathcal{K}| = |\mathcal{J}^< \times \mathcal{K}| = (\frac{|V|^2}{2} - \frac{|V|}{2} - |\tilde{E}_{\leftrightarrow}|) \cdot (|V| + 2|E_{\leftrightarrow}|)$, and it gives an upper bound on $|\mathcal{H}|$, namely $|\mathcal{H}| \leq |\mathcal{J} \otimes \mathcal{K}|$, as selected pairs could be redundant.



Figure 5: An example of two equivalent graphs. Left: G_1 (IV Graph). Right: G_2 .

Example 3.3. Consider an example of showing that $\mathcal{M}(G_1) \subseteq \overline{\mathcal{M}(G_2)}$ in Figure 5. By Theorem 3.1, we have $\mathcal{J}^< = \{(1, 2), (1, 3)\}$, that is the set of $(u, v) \in V^2$ such that $u \leftrightarrow v \notin \mathcal{H}$

$\tilde{G}$, $u < v$. Moreover, $\mathcal{K} = \{(1, 1), (2, 2), (3, 3), (2, 3), (3, 2)\}$, that is the set of $(w_1, w_2) \in V^2$ such that $w_1 = w_2$ or $w_1 \leftrightarrow w_2 \in G$. By definition, we have

$$\begin{aligned} \mathcal{J} \otimes \mathcal{K} = \{ & (1, 1, 2, 1), (1, 1, 3, 1), \\ & (1, 2, 2, 2), (1, 2, 3, 2), \\ & (1, 3, 2, 3), (1, 3, 3, 3), \\ & (1, 2, 2, 3), (1, 2, 3, 3), \\ & (1, 3, 2, 2), (1, 3, 3, 2)\}. \end{aligned}$$

This corresponds with the indices in our quadratic system:

$$\begin{aligned} \mathcal{L} = \{ & A_{11}A_{21} = 0, A_{11}A_{31} = 0, \\ & A_{12}A_{22} = 0, A_{12}A_{32} = 0, \\ & A_{13}A_{23} = 0, A_{13}A_{33} = 0, \\ & A_{12}A_{23} = 0, A_{12}A_{33} = 0, \\ & A_{13}A_{22} = 0, A_{13}A_{32} = 0\}. \end{aligned}$$

It is clear that $\mathcal{L} = 0$ is solvable if and only if we can select one of the two factors for each of the quadratic product so that the selected factors are jointly solvable. As an example, we see that

$$\begin{aligned} \mathcal{L} = \{ & A_{11} \boxed{A_{21}} = 0, A_{11} \boxed{A_{31}} = 0, \\ & \boxed{A_{12}} A_{22} = 0, \boxed{A_{12}} A_{32} = 0, \\ & \boxed{A_{13}} A_{23} = 0, \boxed{A_{13}} A_{33} = 0, \\ & \boxed{A_{12}} A_{23} = 0, \boxed{A_{12}} A_{33} = 0, \\ & \boxed{A_{13}} A_{22} = 0, \boxed{A_{13}} A_{32} = 0\} \\ = \{ & A_{21} = 0, A_{31} = 0, A_{12} = 0, A_{13} = 0 \} \end{aligned}$$

is a solvable system.

Using our newly introduced notation, to characterize the selection of the red marked factors, we can simply define

$$\begin{aligned} \mathbf{S} \subseteq \mathcal{J} \otimes \mathcal{K} = \{ & (1, 2, 2, 2), (1, 2, 3, 2), \\ & (1, 3, 2, 3), (1, 3, 3, 3), \\ & (1, 2, 2, 3), (1, 2, 3, 3), \\ & (1, 3, 2, 2), (1, 3, 3, 2)\} \end{aligned}$$

consisting of the last eight elements in $\mathcal{J} \otimes \mathcal{K}$, as we aim to select the first two indices from them, so that $\mathbf{S}[1] = \{(1, 2), (1, 3)\}$. From the remaining, we select the last two indices, that is $(\mathcal{J} \otimes \mathcal{K} \setminus \mathbf{S})[2] = \{(2, 1), (3, 1)\}$. We therefore obtained $\mathcal{H} = \mathbf{S}[1] \cup (\mathcal{J} \otimes \mathcal{K} \setminus \mathbf{S})[2] = \{(1, 2), (1, 3), (2, 1), (3, 1)\}$, corresponding to the indices of the red marked factors. Moreover, we see that $\mathcal{H}$ induces a linear system.

We now formalize the result:

Lemma 3.3. *Let G and $\tilde{G}$ be two DMGs with the same vertex set V . Let $b_{ij} = (I - \Lambda)_{ij}^{-T}$. Then $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$ if and only if there exists a split $H_S(G_B, \tilde{G}_B)$ so that for any generic $\Lambda \in \mathbb{R}^{G_D}$,*

the following linear system

$$\left\{ \sum_{k \in V \setminus i} b_{kj} \tilde{\lambda}_{ki} = b_{ij} : (i, j) \in \mathcal{H} \right\} \quad (3.16)$$

is solvable in $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}^D}$.

Proof. Using Lemma 3.2, we can write the system $\mathcal{L}$ as:

$$\mathcal{L} = \{f_1 g_1 = 0, f_2 g_2 = 0, \dots, f_m g_m = 0\}, \quad (3.17)$$

where $f_1, \dots, f_m, g_1, \dots, g_m$ are affine linear in $\tilde{\Lambda}$. Use Theorem 3.1, we know that model inclusion holds if and only if $\mathcal{L}$ is generically solvable in $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}^D}$. Let $\tilde{\Lambda}^*$ be such a solution if exists. We define $S \subseteq [m]$ to be the subset that $f_s(\tilde{\Lambda}^*) = 0$ for all $s \in S$. It follows that $g_r(\tilde{\Lambda}^*) = 0$ for all $r \in [m] \setminus S$, and the quadratic system reduces to the linear system $\{f_s = 0 : s \in S\} \cup \{g_r = 0 : r \in [m] \setminus S\}$ in $\tilde{\Lambda}$. The equation $A_{uw_1} \cdot A_{vw_2} = 0$ in $\mathcal{L}$ holds if and only if at least one of its factor is zero. The solvability of the system of equations is thus equivalent to the existence of a partition of the set $\mathcal{J} \otimes \mathcal{K} = \mathbf{S} \cup \mathbf{S}^c$ so that $A_{ij} = 0$ for all (i, j) coming from $\mathbf{S}[1]$ and from $\mathbf{S}^c[2]$. The result follows by noticing that, using Lemma 3.2, $A_{ij} = 0$ if and only if $b_{ij} - \sum_{k \in V \setminus i} b_{kj} \lambda_{ki} = 0$. $\square$

Note that the above theorem can be naturally formulated for exact model equivalence by checking solvability in $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}^D}$ for all $\Lambda \in \mathbb{R}^{G^D}$.

We now make conventions on manipulating submatrices: For a matrix A , let A_{IJ} be the induced submatrix of A by taking rows from I and taking columns from J . Let A_J be the induced submatrix of A by simply taking columns from J . Let $A_{I, \bullet}$ be the induced submatrix by taking rows from I . We now shift our focus to ADMGs. The following lemma shows that model equivalence for ADMGs can be eventually characterized by some rank constraints.

Lemma 3.4. *Let G and $\tilde{G}$ be two ADMGs with the same vertex set V . Let $B = (I - \Lambda)^{-T}$. Then $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$ if and only if there exists a split $H_S(G_B, \tilde{G}_B)$ so that for a generic choice of $\Lambda \in \mathbb{R}^{G^D}$ and all $i \in V$ such that:*

$$\text{rank } B_{(\text{pa}_{\tilde{G}}(i) \cup i)J_i} = \text{rank } B_{\text{pa}_{\tilde{G}}(i)J_i}. \quad (3.18)$$

Here we define $J_i := \{j \in V : (i, j) \in \mathcal{H}\}$.

Proof. First we see that the linear system in Lemma 3.3 is equivalent to $(B_j)^T \tilde{\Lambda}_i = b_{ij}$ for all $(i, j) \in \mathcal{H}$. The generic solvability of $\tilde{\Lambda}$ can be decomposed into generic solvabilities for all $\tilde{\Lambda}_i$ since the systems are independent for distinct columns of $\tilde{\Lambda}$. For a fixed $i \in (i, j)$ where the pair (i, j) comes from $\mathcal{H}$, we define J_i to be the collection of all matched $j \in (i, j)$. The generic solvability amounts to the generic solvability of the canonical linear system $(B_{J_i})^T \tilde{\Lambda}_i = (B_{iJ_i})^T$ for all i choosing from any candidate pair $(i, j) \in \mathcal{H}$. We need to be careful that $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}^D}$ has reduced degrees of freedom in the sense that some entries are set to be zero. We work around this issue by considering the generic solvability of the variable-reduced equivalent system $(B_{I_i J_i})^T \tilde{\Lambda}_{i i} = (B_{i J_i})^T$, where I_i collects the possible nonzero indices $V \setminus \{r \in V : \tilde{\Lambda}_{ri} = 0 \text{ fixed}\}$, that is exactly the set $\{r \in V : r \rightarrow i \in \tilde{G}\} = \text{pa}_{\tilde{G}}(i)$. Now we do not need to worry about the restrictiveness in the domain, and the generic solvability in $\tilde{\Lambda}_i$ simplifies to the rank condition $\text{rank}[(B_{I_i J_i})^T \mid (B_{i J_i})^T] = \text{rank}[(B_{i J_i})^T]$. We finish the proof by noticing that the left side is equal to $\text{rank } B_{(I_i \cup i)J_i}$. $\square$

Note that we didn't claim the same for DMGs with directed cycles. The reason for it is rather subtle. The lemma above is based on Theorem 3.1, where we turned the parameterization problem into the solvability of a system of quadratic equations. It is important that the system $\mathcal{L}$ should be solvable in $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$, where the condition $\det(I - \tilde{\Lambda}) \neq 0$ is implicitly included by definition of the domain $\mathbb{R}^{\tilde{G}_D}$. However, in Lemma 3.4, by rewriting the solvability condition into a rank condition, the restriction on the domain can no more be captured. This drawback comes naturally with the rank formulation as such a nonlinear constraint on the determinant cannot simply be captured by rank conditions. We illustrate this caveat with the following example.

Example 3.4. For a cyclic case, let $G, \tilde{G}$ be defined as in Figure 6. To check whether $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$, we first use Theorem 3.7 and verify that all path rank conditions hold when choosing the vertex cover $Q = \{(2, 1), (3, 1), (1, 2), (1, 3)\}$ for $G_{\mathcal{I}}$. However, the constraint on the determinant fails. To see this, we can use Theorem 3.1 and compute

$$A := \begin{pmatrix} 1 & 0 & 0 \\ -\lambda_{12}\lambda_{23}\tilde{\lambda}_{32} + \lambda_{12} & \bullet & \bullet \\ \lambda_{12}\lambda_{23} - \lambda_{12}\tilde{\lambda}_{23} & \bullet & \bullet \end{pmatrix}. \quad (3.19)$$

Next, we verify that $\mathcal{L} = 0$ simplifies to requiring $A_{21} = 0$ and $A_{31} = 0$. This implies that solutions to $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$ have the form $\tilde{\lambda}_{32} = \frac{1}{\lambda_{23}}$ and $\tilde{\lambda}_{23} = \lambda_{23}$. But then $\det(I - \tilde{\Lambda}) = 1 - \tilde{\lambda}_{23}\tilde{\lambda}_{32} = 1 - \frac{\lambda_{23}}{\lambda_{23}} = 0$ for almost all $\Lambda \in \mathbb{R}^{G_D}$. So we conclude that there is indeed no generic solution to $\tilde{\Lambda} \in \mathbb{R}_{reg}^{\tilde{G}_D}$, and $\mathcal{M}(G) \not\subseteq \overline{\mathcal{M}(\tilde{G})}$.



Figure 6: An example illustrating the necessity of checking the determinant for DMGs with cycles. Let the DMG on the left hand side be G and the one on the right hand side be $\tilde{G}$.

Nevertheless, this rank approach is essential as it allows for a fully graphical result for acyclic cases. In the following, we introduce the well known path rank which provides a combinatorial view of the rank conditions. Notice that some results below are stated for DMGs with cycles.

Definition 3.2 (Path ranks). *In a DMG G , for two sets of vertices $X, Y \subseteq V$, the path rank $r_G(X, Y)$ is the maximum number of non-intersecting directed paths from X to Y in G_D .*

Two paths are **non-intersecting** if they are vertex-disjoint. This max-flow quantity has an equivalent min-cut version, which is defined to be the minimum number of nodes removal $|Z|$ from G to ensure no directed path from $X \setminus Z$ to $Y \setminus Z$ [Menger, 1927]. In other words, path ranks summarize the size of “bottlenecks” [Bing et al., 2026]. The following lemma provides a link between matrix rank and path rank [Lindström, 1973, Gessel and Viennot, 1985, Talaska, 2012].

Lemma 3.5. *In a DMG G , for any two sets of vertices $Z, Y \subseteq V$ that need not be disjoint, the following equality holds for $B = (I - \Lambda)^{-T}$ induced by a generic choice of $\Lambda \in \mathbb{R}^{G_D}$:*

$$\text{rank } B_{XY} = r_G(Y, X). \quad (3.20)$$

Equipped with all the necessary preparations, we are now close to our first graphical characterization. The only remaining question is whether the confounding split admits a graphical

characterization. This motivates our next definition of the induced graph, which elegantly bridges the sequence of binary selections and the vertex cover problem.

Definition 3.3 (Induced graph). *Let G and $\tilde{G}$ be two DMGs with the same vertex set V . Let $G_{\mathcal{I}}(G_B, \tilde{G}_B) = (V \times V, E_{\mathcal{I}})$ be the induced undirected graph with its vertices indexed by pairs, and $\{(u, w_1), (v, w_2)\} \in E_{\mathcal{I}}$ if $(u, v) \in \mathcal{J}$ and $(w_1, w_2) \in \mathcal{K}$.*

For an undirected graph G , a **vertex cover** is a set of vertices $Q \subseteq V$ that includes at least one endpoint of every edge of the graph. This concept connects confounding split in the following way:

Lemma 3.6. *Let G and $\tilde{G}$ be two DMGs with the same vertex set. Each confounding split $\mathcal{H}$ is a vertex cover of $G_{\mathcal{I}}(G_B, \tilde{G}_B)$. Moreover, each vertex cover Q contains a confounding split $\mathcal{H}$.*

Proof. Assume $\mathcal{H} = H_S(G_B, \tilde{G}_B)$ is a confounding split, then for each $(u, w_1, v, w_2) \in \mathcal{J} \otimes \mathcal{K}$, either (u, w_1) or (v, w_2) is in $\mathcal{H}$. Since each edge in $G_{\mathcal{I}}(G_B, \tilde{G}_B)$ has endpoints (u, w_1) and (v, w_2) , where $(u, w_1, v, w_2) \in \mathcal{J} \otimes \mathcal{K}$, we conclude that $\mathcal{H}$ must include one of these endpoints. Assume now that Q is a vertex cover of $G_{\mathcal{I}}$, then for each $(u, w_1, v, w_2) \in \mathcal{J} \otimes \mathcal{K}$, at least one of the pairs (u, w_1) and (v, w_2) must be in Q . For each $(u, w_1, v, w_2) \in \mathcal{J} \otimes \mathcal{K}$, if $(u, w_1) \in Q$, we let $(u, w_1, v, w_2) \in \mathbf{S}$. It is clear that for this particular $S \subseteq \mathcal{J} \otimes \mathcal{K}$ we have $H_S(G_B, \tilde{G}_B) \subseteq Q$. $\square$

We now arrive at our first graphical characterization on model equivalence:

Theorem 3.7. *Let G and $\tilde{G}$ be two ADMGs with the same vertex set V . Then $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$ if and only if there exists a vertex cover $Q \subseteq V^2$ of the induced graph $G_{\mathcal{I}}(G_B, \tilde{G}_B)$ such that for all $i \in V$:*

$$r_G(Q_i, \text{pa}_{\tilde{G}}(i)) = r_G(Q_i, \text{pa}_{\tilde{G}}(i) \cup i). \quad (3.21)$$

Here we define $Q_i := \{j \in V : (i, j) \in Q\}$.

Proof. Applying Lemma 3.5 to Lemma 3.4, adopting the notations there, we conclude that $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ if and only if there exists a confounding split $\mathcal{H}$ such that for all i with $(i, j) \in \mathcal{H}$: $r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) = r_G(J_i, \text{pa}_{\tilde{G}}(i))$. Note that the J_i in the earlier result is the same as Q_i in this theorem, we write Q_i to emphasize the connection with the vertex cover Q . The necessary direction follows from Lemma 3.6 that each confounding split is also a vertex cover of the induced graph. For the sufficient direction, assume there is a vertex cover Q such that the corresponding path rank condition holds. By Lemma 3.6, we know that there exists a confounding split $\mathcal{H} \subseteq Q$ such that for all i with $(i, j) \in \mathcal{H}$: $r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) = r_G(J_i, \text{pa}_{\tilde{G}}(i))$. Therefore, $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$. We finish the proof by noticing that the condition of choosing i from a pair $(i, j) \in Q$ can be dropped, because otherwise, $J_i = \emptyset$ and $r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) = r_G(J_i, \text{pa}_{\tilde{G}}(i)) = 0$ always hold. $\square$

Example 3.5. Consider again the example of showing $\mathcal{M}(G_1) \subseteq \overline{\mathcal{M}(G_2)}$ with G_1, G_2 as defined in Figure 7. The induced graph is shown in Figure 8a. Each quadratic product $A_{vw_1}A_{uw_2} \in \mathcal{L}$ in the system $\mathcal{L}$ is represented as an undirected edge in the induced graph. To show model inclusion, we need to find a vertex cover of $G_{\mathcal{I}}$ satisfying all path rank constraints. We show that $Q = \{(2, 1), (3, 1), (1, 2), (1, 3)\} \subseteq V^2 = V(G_{\mathcal{I}})$ is one of such. First, it is easy to see that Q covers all edges in $G_{\mathcal{I}}$. Next, we verify that

$$\begin{aligned} i = 1, \quad Q_1 &= \{2, 3\}, \quad \text{pa}_{G_2}(1) = \emptyset, \quad r_{G_1}(Q_1, \emptyset) = r_{G_1}(Q_1, \{1\}) = 0 \quad \checkmark \\ i = 2, \quad Q_2 &= \{1\}, \quad \text{pa}_{G_2}(2) = \{1\}, \quad r_{G_1}(Q_2, \{1\}) = r_{G_1}(Q_2, \{1, 2\}) = 1 \quad \checkmark \\ i = 3, \quad Q_3 &= \{1\}, \quad \text{pa}_{G_2}(3) = \{1\}, \quad r_{G_1}(Q_3, \{1\}) = r_{G_1}(Q_3, \{1, 3\}) = 1 \quad \checkmark \end{aligned}$$

Table 2: A comparison of confounding split and vertex cover.

	Confounding split ($\mathcal{H}$)	Vertex cover (Q)
Notation	$\mathcal{H} = H_S(G_B, \tilde{G}_B) \subseteq V^2$	$Q \subseteq V(G_I) = V^2$
Induced by	$G_B, \tilde{G}_B$	$G_B, \tilde{G}_B$
Operation	binary selection from products	vertex selection
Operates on	system of quadratic equations	induced graph
Viewpoint	algebraic	combinatorial
Relation to each other	is a vertex cover	contains a confounding split

We therefore conclude $\mathcal{M}(G_1) \subseteq \overline{\mathcal{M}(G_2)}$.



Figure 7: An example of two equivalent graphs. Left: G_1 (IV Graph). Right: G_2 .

There exists an equivalent local characterization in terms of the so-called edge rank [Dai et al., 2026] (or matching rank), which directly operates on edges instead of paths.

Definition 3.4 (Edge ranks/Matching ranks). *In a DMG G , for two sets of vertices $X, Y \subseteq V$, the edge rank (or matching rank) $m_G(X, Y)$ is the size of the maximum bipartite matching from X to Y via edges in G_D , where self-matches (v to v if $v \in X \cap Y$) are allowed.*

Edge ranks admit the following min-cut version:

$$m_G(X, Y) := \min_{\substack{X' \subseteq X, Y' \subseteq Y, \\ X \cap Y \subseteq X' \cup Y'}} \{|X'| + |Y'| : \text{there is no edge from } X \setminus X' \text{ to } Y \setminus Y' \text{ in } G_D\}. \quad (3.22)$$

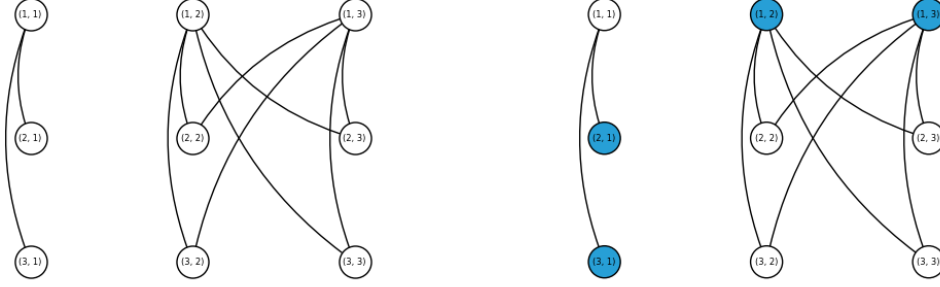
There is a duality between edge ranks and path ranks, which has been studied in the matroid setting [Perfect, 1968, Ingleton and Piff, 1973] and has recently received attention in graphical models:

Lemma 3.8 ([Dai et al., 2026]). *In a DMG with vertices in V , for any $X, Y \subseteq V$ that need not be disjoint, the following equality holds:*

$$\min(|X|, |Y|) - r_G(X, Y) = |V| - \max(|X|, |Y|) - m_G(V \setminus Y, V \setminus X). \quad (3.23)$$

It has also been shown that, while path ranks operate on the mixing matrix $(I - \Lambda)^{-T}$, edge ranks operate directly on the directed edge matrix Λ :

Lemma 3.9 ([cf. Dai et al., 2026]). *The edge rank $m_G(X, Y)$ is equal to the maximum number of nonzero entries that can be positioned on the diagonal by permuting the columns (or rows) of*



(a) The Induced graph $G_I(G_1, G_2)$ with G_1, G_2 as in Figure 7. (b) A vertex cover satisfying all path rank constraints.

Figure 8: An Induced graph with its vertex cover.

the submatrix $\hat{\Lambda}_{XY}$, where the entries of $\hat{\Lambda} \in \{0, 1\}^{V \times V}$ is defined by

$$\hat{\Lambda}_{ij} = \begin{cases} 1 & \text{if } i = j \text{ or } i \rightarrow j \in G_D, \text{ and} \\ 0 & \text{otherwise.} \end{cases} \quad (3.24)$$

In an undirected graph G with vertices in V , we say $W \subseteq V$ is **independent** if there is no edge between any pair of vertices in W [Godsil and Royle, 2001, p. 3]. It is straightforward to see that this concept is directly connected with vertex cover in the following way:

Lemma 3.10. *For an undirected graph G with its vertices in V , the following holds:*

$$Q \subseteq V \text{ is a vertex cover of } G \iff V \setminus Q \text{ is independent in } G. \quad (3.25)$$

Proof. For the “ $\Leftarrow$ ” direction, if $W \subseteq V$ is independent in G , then all edges in G must either have both endpoints in $V \setminus W$ or exactly one endpoint in $V \setminus W$. We conclude that $V \setminus W$ is a vertex cover of G . For the “ $\Rightarrow$ ” direction, $Q \subseteq V$ is a vertex cover of G implies that there is no edge using both endpoints in $V \setminus Q$, and we are done. $\square$

To simplify notations, let us write $\text{npa}_G(i) := V \setminus \text{pa}_G(i)$ for the set of non-parents of i in G_D . Now we obtain the following dual graphical characterization:

Theorem 3.11. *Let G and $\tilde{G}$ be two ADMGs with the same vertex set V . Then $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$ if and only if there exists an independent set $W \subseteq V^2$ of the induced undirected graph $G_I(G_B, \tilde{G}_B)$ such that for all $i \in V$:*

$$m_G(\text{npa}_{\tilde{G}}(i), W_i) = m_G(\text{npa}_{\tilde{G}}(i) \setminus i, W_i) + 1. \quad (3.26)$$

Here we define $W_i := \{j : (i, j) \in W\}$.

Proof. We start from rewriting the path rank characterization $r_G(Q_i, \text{pa}_{\tilde{G}}(i)) = r_G(Q_i, \text{pa}_{\tilde{G}}(i) \cup i)$ from Theorem 3.7 using Lemma 3.8. Let $X := Q_i$ and $Y := \text{pa}_{\tilde{G}}(i)$. Since

$$r_G(X, Y) = \min(|X|, |Y|) - |V| + \max(|X|, |Y|) + m_G(V \setminus Y, V \setminus X), \quad (3.27)$$

the path rank condition is equivalent to

$$\begin{aligned} & \min(|X|, |Y|) + \max(|X|, |Y|) + m_G(V \setminus Y, V \setminus X) \\ &= \min(|X|, |Y \cup i|) + \max(|X|, |Y \cup i|) + m_G(V \setminus (Y \cup i), V \setminus X). \end{aligned} \quad (3.28)$$

If $|X| \leq |Y|$, the above equation simplifies to

$$\begin{aligned} & |X| + |Y| + m_G(V \setminus Y, V \setminus X) \\ &= \underbrace{|X|}_{\min(|X|, |Y \cup i|)} + \underbrace{|Y| + 1}_{\max(|X|, |Y \cup i|)} + m_G(V \setminus (Y \cup i), V \setminus X), \end{aligned} \quad (3.29)$$

as $i \notin Y$. Otherwise, $|X| > |Y|$ implies $|X| \geq |Y| + 1$ and Equation (3.28) becomes

$$\begin{aligned} & |Y| + |X| + m_G(V \setminus Y, V \setminus X) \\ &= \underbrace{|Y| + 1}_{\min(|X|, |Y \cup i|)} + \underbrace{|X|}_{\max(|X|, |Y \cup i|)} + m_G(V \setminus (Y \cup i), V \setminus X). \end{aligned} \quad (3.30)$$

In all the cases, we obtain

$$m_G(V \setminus Y, V \setminus X) = m_G(V \setminus (Y \cup i), V \setminus X) + 1, \quad (3.31)$$

which is by definition

$$m_G(\text{npa}_{\tilde{G}}(i), V \setminus Q_i) = m_G(\text{npa}_{\tilde{G}}(i) \setminus i, V \setminus Q_i) + 1. \quad (3.32)$$

We finish the proof by expressing the complement of vertex cover as an independent set using Lemma 3.10. $\square$

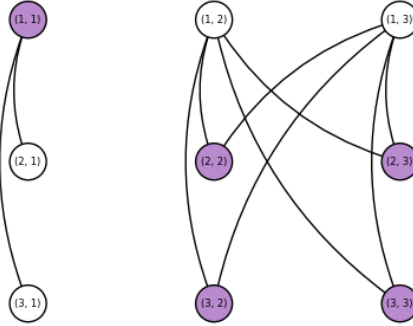


Figure 9: An independent set.

Example 3.6. Consider again the example of showing $\mathcal{M}(G_1) \subseteq \overline{\mathcal{M}(G_2)}$ with G_1, G_2 as defined in Figure 7. Using the dual theorem, to show model inclusion, we need to find an independent set of $G_{\mathcal{I}}$ satisfying all matching rank constraints. We show that $W =$

Table 3: Dual characterizations.

	Theorem 3.7	Theorem 3.11
Rank constraints	path ranks	matching (edge) ranks
Corresponding matrix ranks	mixing matrix $(I - \Lambda)^{-T}$	directed edge matrix Λ
Graphical representation	non-intersecting paths	bipartite matching
Confounding characterizations	vertex cover	independent set

$\{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3)\} \subseteq V^2 = V(G_{\mathcal{I}})$ is one of such (Figure 9). First, it is easy to see that there is no edge between vertices in W in $G_{\mathcal{I}}$. Next, we verify that

$$\begin{aligned}
i = 1, W_1 = \{1\}, \text{npa}_{G_2}(1) = \{1, 2, 3\}, m_{G_1}(\{1, 2, 3\}, W_1) = 1, m_{G_1}(\{2, 3\}, W_1) = 0 & \quad \checkmark \\
i = 2, W_2 = \{2, 3\}, \text{npa}_{G_2}(2) = \{2, 3\}, m_{G_1}(\{2, 3\}, W_2) = 2, m_{G_1}(\{3\}, W_2) = 1 & \quad \checkmark \\
i = 3, W_3 = \{2, 3\}, \text{npa}_{G_2}(3) = \{2, 3\}, m_{G_1}(\{2, 3\}, W_3) = 2, m_{G_1}(\{2\}, W_3) = 1 & \quad \checkmark
\end{aligned}$$

We therefore conclude $\mathcal{M}(G_1) \subseteq \overline{\mathcal{M}(G_2)}$.

3.2 Towards a Simpler Characterization

Theorem 3.7 and Theorem 3.11 provide the first combinatorial answers to model equivalence. However, to show $\mathcal{M}(G) \not\subseteq \overline{\mathcal{M}(\tilde{G})}$, one needs to go through all possible vertex cover of the induced graph (equivalently, the dual characterization requires checking all independent sets), which is practically infeasible. To simplify this problem, we develop the following consistency tools to analyze the maximal set J_i that can be chosen such that the path rank condition (3.21) still holds. This amounts to reformulating the problem of “finding a suitable vertex cover ensuring the path rank conditions” into the problem of “optimizing J_i subject to the path rank conditions to obtain a valid vertex cover”. We now turn our focus to Equation (3.21), which is the central equality governing when a vertex $i \in V$ can be added to the target set without increasing the path rank. To this end, we define the following:

Definition 3.5 (Consistency). *Let $G = (V, E)$ be a DAG. Let $X, Y \subseteq V$ be two subsets of nodes that need not be disjoint, and $i \in V$ be a vertex that is not in Y . We say that the triple (X, Y, i) is **consistent** (or simply that X is **consistent**, since Y and i are typically fixed and X is the only variable) if $r_G(X, Y) = r_G(X, Y \cup \{i\})$, and **inconsistent** otherwise.*

It is clear that $r_G(X, Y) \leq r_G(X, Y \cup i)$ holds always. Moreover, the inequality appears if and only if $r_G(X, Y) + 1 = r_G(X, Y \cup i)$ since the path rank difference can be maximal one. In the context of Theorem 3.7, the set $Y \subseteq V$ and vertex $i \in V$ are both given. To determine how to choose a feasible $X \subseteq V$ such that the triple (X, Y, i) remains consistent, we need to understand which nodes can always be included and which can never be included. We give answers to both.

Lemma 3.12. *Let $G = (V, E)$ be a DAG. Let $X, Y \subseteq V$ be two subsets that need not be disjoint, and $i \in V$ be a vertex that is not in Y . Let $X_0 = V \setminus \text{an}(Y \cup i)$. If (X, Y, i) is consistent, then the extension $(X \cup X_0, Y, i)$ is also consistent.*

Proof. Given a fixed consistent triple (X, Y, i) , assume that $r_G(X, Y) = r_G(X, Y \cup i) = k$. For $x \in X_0$, we see that x has no directed path to $Y \cup i$, and therefore also no directed path to Y , thus $X_0 = \{x \in V : r_G(x, Y \cup i) = 0\}$. Consider the min-cut version of the path rank condition and let $B \subseteq V$ be the minimal blocking set for (X, Y) and $(X, Y \cup i)$ with $|B| = k$, then B blocks also $(X \cup X_0, Y)$ and $(X \cup X_0, Y \cup i)$. As a result, we have $r_G(X \cup X_0, Y) = r_G(X \cup X_0, Y \cup i) = k$. $\square$

Lemma 3.13. *Let $G = (V, E)$ be a DAG. Let $X, Y \subseteq V$ be two subsets that need not be disjoint, and $i \in V$ be a vertex that is not in Y . Let $x_0 \in \text{an}(i) \setminus \text{an}(Y)$. Then every triple (X, Y, i) with X containing such x_0 is inconsistent.*

Proof. For $x_0 \in \text{an}(i) \setminus \text{an}(Y)$, we first notice that (x_0, Y, i) is inconsistent, as $r_G(x_0, Y \cup i) = 1$ and $r_G(x_0, Y) = 0$. Assume for contradiction that (X, Y, i) is consistent with some $X \subseteq V$ containing x_0 , i.e., $r_G(X, Y) = r_G(X, Y \cup i) = k$ with $x_0 \in X$. Let $X_0 = \text{an}(i) \setminus \text{an}(Y)$ be the collection of such $x_0 \in V$. It follows that $r_G(X \setminus X_0, Y) = k$. Further, $r_G(X \setminus X_0, Y \cup i) < k$ as we closed all paths from X_0 to i that do not intersect with any other paths from $X \setminus X_0$ to Y . As a result, the path rank must drop by one. This is a contradiction to $r_G(X \setminus X_0, Y) \leq r_G(X \setminus X_0, Y \cup i)$. $\square$

The following lemma claims that every “subset” of a consistent triple is consistent.

Lemma 3.14. *Let $G = (V, E)$ be a DAG. Let $X, Y \subseteq V$ be two subsets that need not be disjoint, and $i \in V$ be a vertex that is not in Y . Assume (X, Y, i) is consistent, then (S, Y, i) is consistent for all $S \subseteq X$.*

Proof. Assume for contradiction that (X, Y, i) is consistent with $S \subseteq X$ such that (S, Y, i) is inconsistent. Assume $r_G(S, Y) = k$ and $r_G(S, Y \cup i) = k + 1$. By construction, there must exist a step by step extension $S_{k+1} = S_k \cup x_{k+1}$ with $k \in \{0, 1, \dots, |X \setminus S| - 1\}$ and $x_{k+1} \in X \setminus S_k$ such that $S_0 = S$ and $S_{|X \setminus S|} = X$. Assume for some k that $r_G(S_{k+1}, Y) - r_G(S_k, Y) = 1$, meaning that adding the vertex x_{k+1} increases the bottleneck to Y , then this increases also the bottleneck to $Y \cup i$, and $r_G(S_{k+1}, Y \cup i) - r_G(S_k, Y \cup i) = 1$. Therefore, it is impossible to add a vertex that increases the bottleneck to Y while leaving the bottleneck to $Y \cup i$ unchanged. Namely, $r_G(S_k, Y \cup i) - r_G(S_k, Y) \geq 1$ for all k . This contradicts (S_k, Y, i) is consistent for $k = |X \setminus S| - 1$. $\square$

Corollary 3.15. *Let $G = (V, E)$ be a DAG. Let $X, Y \subseteq V$ be two subsets that need not be disjoint, and $i \in V$ be a vertex that is not in Y . Assume (X, Y, i) is inconsistent, then no (S, Y, i) can be consistent with S containing X .*

Proof. Assume (X, Y, i) is inconsistent with (S, Y, i) being consistent and $S \supseteq X$. This contradicts Lemma 3.14 because X would be an inconsistent subset of S . $\square$

The above corollary indicates that once we reach an inconsistent state, it is no longer possible to recover consistency by simply adding more vertices to the source set.

Lemma 3.16. *Let $G = (V, E)$ be a DAG. Let $Y \subseteq V$ be a subset and $i \in V$ be a vertex that is not in Y . Then the triple $(V \setminus \text{an}(i), Y, i)$ is consistent.*

Proof. Let $S = \text{an}(Y) \setminus \text{an}(i)$. We first show that the triple (S, Y, i) is consistent. Assume $r_G(S, Y) = k$, then $r_G(S, Y \cup i) = k$ holds as well since a minimal blocking set for (S, Y) blocks also $(S, Y \cup i)$. Apply Lemma 3.12 we realize that

$$(S \cup (V \setminus \text{an}(Y \cup i)), Y, i) = ((\text{an}(Y) \setminus \text{an}(i)) \cup (V \setminus \text{an}(Y \cup i)), Y, i) \quad (3.33)$$

$$= (V \setminus \text{an}(i), Y, i) \quad (3.34)$$

is consistent too. $\square$

In an undirected graph G , for $v \in V$, we define the **neighbors** of v to be $\text{ne}_G(v) := \{w \in V \setminus v : v - w \in G\}$. For a set of vertices $S \subseteq V$, we define the neighbors of S simply by the union $\text{ne}_G(S) := \cup_{v \in S} \text{ne}_G(v)$. For any $S \subseteq V$, we define the **projected graph** G^S to be subgraph of G which has vertices in S and edges with both endpoints in S .

Definition 3.6 (Directly decidable). *Let G and $\tilde{G}$ be two ADMGs with the same vertex set V . Let*

$$X = \{(i, j) \in V^2 : j \in \text{an}_G(\text{pa}_{\tilde{G}}(i) \cup i)\} \quad (3.35)$$

*be the **relevant set**. Next, let*

$$S = \{(i, j) \in X : j \in \text{an}_G(i) \setminus \text{an}_G(\text{pa}_{\tilde{G}}(i))\} \quad (3.36)$$

*be the **unselectable set**. Let $G_{\mathcal{I}} := G_{\mathcal{I}}(G_B, \tilde{G}_B)$. The pair $(G, \tilde{G})$ is called **directly decidable** if $\text{ne}_{G_{\mathcal{I}}}^X(S)$ is a vertex cover of $G_{\mathcal{I}}^X$ or there is an edge in $G_{\mathcal{I}}^X$ with both endpoints in S .*

The above graphical definition connects the model parametrization and the algebraic result (Theorem 3.1) in the following way, revealing intrinsic properties of the vertices in the relevant and unselectable sets:

Lemma 3.17. *Consider two ADMGs $G, \tilde{G}$ with their directed edges parametrized by $\Lambda, \tilde{\Lambda}$. Let $A = (I - \tilde{\Lambda})^T(I - \Lambda)^{-T}$ be the mixing matrix, and let X, S be the relevant set and the unselectable set as defined in (3.35) and (3.36), then*

1. $(i, j) \in X \iff A_{ij} \neq 0$,
2. $(i, j) \in S \iff A_{ij}$ is nonzero and does not depend on $\tilde{\Lambda}$.

Proof. As we have already shown, $A_{ij} = b_{ij} - \sum_{k \in V \setminus i} \tilde{\lambda}_{ki} b_{kj}$. To prove the first claim, for the “if” direction, we show that $j \notin \text{an}_G(\text{pa}_{\tilde{G}}(i) \cup i)$ implies $A_{ij} = 0$. To begin with, we notice that $j \notin \text{an}_G(\text{pa}_{\tilde{G}}(i) \cup i)$ implies that j is not an ancestor of i in G and also not an ancestor of $\text{pa}_{\tilde{G}}(i)$ in G . The first point already implies that $b_{ij} = 0$ using the properties of the path matrix. It remains to show that $\sum_{k \in V \setminus i} \tilde{\lambda}_{ki} b_{kj} = 0$. This follows the fact that if $\tilde{\lambda}_{ki}$ is nonzero, then $b_{kj} = 0$ as $j \notin \text{an}_G(\text{pa}_{\tilde{G}}(i))$. For the “only if” direction, if $j \in \text{an}_G(\text{pa}_{\tilde{G}}(i) \cup i)$, then either $b_{ij} \neq 0$ or there exists a $k \in \text{pa}_{\tilde{G}}(i)$ such that j is ancestor of k in G . This implies for such k that $\tilde{\lambda}_{ki} b_{kj} \neq 0$.

We now consider the second claim. For the direct implication, if j is an ancestor of i and not an ancestor of $\text{pa}_{\tilde{G}}(i)$ in G , then b_{ij} is a nonzero constant. It remains to show that $b_{kj} = 0$ whenever $\tilde{\lambda}_{ki}$ is nonzero. This follows directly from the fact that j is not an ancestor of $\text{pa}_{\tilde{G}}(i)$ in G . For the reversed direction, we show that $j \notin \text{an}_G(i) \setminus \text{an}_G(\text{pa}_{\tilde{G}}(i))$ implies that A_{ij} is either zero or depends on $\tilde{\Lambda}$. To begin with, we notice that $j \notin \text{an}_G(i) \setminus \text{an}_G(\text{pa}_{\tilde{G}}(i))$ implies that j is either not an ancestor of i in G or j is an ancestor of $\text{pa}_{\tilde{G}}(i)$ in G . If j is not an ancestor of i in G , then $b_{ij} = 0$ and A_{ij} reduces to a linear function $\sum_{k \in V \setminus i} \tilde{\lambda}_{ki} b_{kj}$ in $\tilde{\Lambda}$. If j is an ancestor of $\text{pa}_{\tilde{G}}(i)$ in G , then there must exist a $k \in V \setminus i$ such that both $\tilde{\lambda}_{ki}$ and b_{kj} are nonzero. $\square$

It is convenient that the exponential complexity of Theorem 3.7 arises from checking all possible vertex covers of the induced graph. However, if a pair of ADMGs is directly decidable, it suffices to check a single vertex cover. The key idea is that, by exploiting the consistency results for the path ranks, we can identify vertices in the induced graph that can always be included in the vertex cover. These vertices are therefore irrelevant to the choice of the vertex cover, allowing us to project the induced graph onto its relevant part, denoted by $G_{\mathcal{I}}^X$. Conversely,

there are vertices that can never be included, as doing so would introduce an imbalance in the path-rank constraints. Consequently, the neighbors of these inconsistent vertices in $G_{\mathcal{I}}^X$ must be included as this is necessary for a valid vertex cover. Finally, if the unselectable neighbors already result a vertex cover, then it is sufficient to check this single candidate. Indeed, if this inclusion-minimal vertex cover fails to satisfy the path-rank constraints, then no larger vertex cover can satisfy them either. We now formalize this result.

Theorem 3.18. *Let G and $\tilde{G}$ two ADMGs. If $\text{ne}_{G_{\mathcal{I}}^X}(S)$ is a vertex cover of $G_{\mathcal{I}}^X$, then $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$ if and only if*

$$r_G(J_i, \text{pa}_{\tilde{G}}(i)) = r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) \quad \text{for all } i \in V. \quad (3.37)$$

Here we define $J_i := \{j \in V : (i, j) \in \text{ne}_{G_{\mathcal{I}}^X}(S)\}$, where S is the unselectable set as defined in (3.36).

Proof. We first claim that Theorem 3.7 can be restricted on a subgraph in the following way, namely, $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$ if and only if there exists a vertex cover V' of $G_{\mathcal{I}}^X$ such that for all $i \in V$: $r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) = r_G(J_i, \text{pa}_{\tilde{G}}(i))$ with $J_i := \{j : (i, j) \in V'\}$.

To show the “if” direction of the claim, let $X_0 = V_{\mathcal{I}} \setminus X$ be the irrelevant set of vertices in $G_{\mathcal{I}}(G_B, \tilde{G}_B)$. If V' is a vertex cover of $G_{\mathcal{I}}^X$, then $V' \cup X_0$ must be a vertex cover of $G_{\mathcal{I}}$ since $G_{\mathcal{I}}^X$ is obtained by removing X_0 and all edges with vertices in X_0 . By using Lemma 3.12, we see that $r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) = r_G(J_i, \text{pa}_{\tilde{G}}(i))$ with $J_i := \{j : (i, j) \in V' \cup X_0\}$ holds after extending V' to $V' \cup X_0$. Therefore, we can always find a suitable vertex cover $V' \cup X_0$ of $G_{\mathcal{I}}$ satisfying all the path rank constraints. To show the “only if” direction of the claim, let V'' be a suitable vertex cover of $G_{\mathcal{I}}$ satisfying all path rank constraints. It is clear that $V'' \cap X$ must be a vertex cover of $G_{\mathcal{I}}^X$. Moreover, $r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) = r_G(J_i, \text{pa}_{\tilde{G}}(i))$ with $J_i := \{j : (i, j) \in V'' \cap X\}$ holds by applying Lemma 3.14, and we conclude $V' := V'' \cap X$ is a suitable vertex cover for $G_{\mathcal{I}}^X$.

Now we move to the main part of the proof. We conclude the sufficient direction by applying the claim on subgraph restriction as proved above. To show the necessary direction, using the subgraph restriction, assume V' is a suitable vertex cover for $G_{\mathcal{I}}^X$ satisfying all path rank constraints, such that for all $i \in V$: $r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) = r_G(J_i, \text{pa}_{\tilde{G}}(i))$ with $J_i := \{j : (i, j) \in V'\}$. By Lemma 3.13, we must have $S \cap V' = \emptyset$ to ensure the consistency. As a result, $\text{ne}_{G_{\mathcal{I}}^X}(S) \subseteq V'$ since every edge which has an endpoint in S must be covered by a vertex in $\text{ne}_{G_{\mathcal{I}}^X}(S)$ as there is no edge with both endpoints in S . We then use Lemma 3.14 and obtain that $r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) = r_G(J_i, \text{pa}_{\tilde{G}}(i))$ with $J_i := \{j : (i, j) \in \text{ne}_{G_{\mathcal{I}}^X}(S)\}$. To see that there is no edge with both endpoints in S , assume for contradiction, that such an edge exists. To ensure a vertex cover, we must include one vertex in S to the vertex cover V' . However, such a vertex cover cannot satisfy all paths rank constraints as shown in Lemma 3.13. $\square$

Corollary 3.19. *If there exists $i \in V$ so that*

$$r_G(J_i, \text{pa}_{\tilde{G}}(i)) \neq r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) \quad (3.38)$$

with $J_i := \{j \in V : (i, j) \in \text{ne}_{G_{\mathcal{I}}^X}(S)\}$. Then $\mathcal{M}(G) \not\subseteq \overline{\mathcal{M}(\tilde{G})}$.

Proof. This follows directly from the fact that taking the neighbors of the unselectable set is necessary for any vertex cover, and if the path rank conditions fail, by Corollary 3.15, we know no larger vertex cover can satisfy them. $\square$

Proposition 3.20. *For a directly decidable pair of ADMGs, model inclusion can be determined in $\mathcal{O}(|V|^4)$.*

Proof. By definition, a pair $(G, \tilde{G})$ is called directly decidable if $\text{ne}_{G_{\mathcal{I}}^X}(S)$ is a vertex cover of $G_{\mathcal{I}}^X$, or there is an edge in $G_{\mathcal{I}}^X$ with both endpoints in S . Assume $\text{ne}_{G_{\mathcal{I}}^X}(S)$ to be a vertex cover of $G_{\mathcal{I}}^X$, then Theorem 3.18 determines model inclusion by checking a single candidate set of path-rank constraints, yielding a worst-case complexity of $\mathcal{O}(|V|^4)$. This follows from the fact that the maximum flow algorithm runs in $\mathcal{O}(|V|^3)$ time [Chen et al., 2022, Tramontano et al., 2026], and this is invoked at most once for each vertex. Notice that ancestors and parents can be determined in $\mathcal{O}(|V|^2)$, and we need to check no more than $\mathcal{O}(|V|^2)$ of them. This preparation step (also in $\mathcal{O}(|V|^4)$) can be done before starting running the maximum flow algorithm. For the other case, assume that an uncovered edge exists between the vertices that cannot be taken. Then we can directly conclude non-equivalence. $\square$

A simulation suggests that the proposed decidability condition (Definition 3.6) is empirically a relatively weak requirement: for small graphs with fewer than ten vertices, conditioning on the number of bidirected edges, we find that around 95% of sampled pairs are directly decidable (see Figure 10). To illustrate the decidability condition, we show induced graphs with the unselectable set S and its neighbors in various cases (Figure 11).

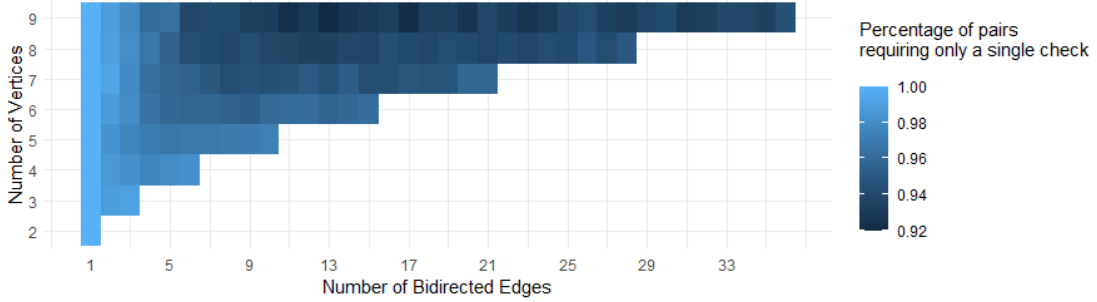


Figure 10: Empirical percentage of ADMG pairs requiring only a single check.

But what can we do if the decidability conditions Theorem 3.18 fail? We first observe that it is always beneficial to consider an induced graph reduction. To introduce this, we write $\text{cl}_G(S) := \text{ne}_G(S) \cup S$ for the closure of S in an undirected graph G . We perform the following:

Definition 3.7 (Reduced induced graph). *Let $G_{\mathcal{I}} = (V^2, E_{\mathcal{I}})$ be an induced graph. Let $X, S \subseteq V^2$ denote the relevant and unselectable sets, respectively, as defined in (3.35) and (3.36). We define the **reduced induced graph** (or simply **reduced graph**) by*

$$G_{\mathcal{I}}^* := G_{\mathcal{I}}^{X \setminus \text{cl}_{G_{\mathcal{I}}}(S)}, \quad (3.39)$$

that is, the projection of $G_{\mathcal{I}}$ onto $X \setminus \text{cl}_{G_{\mathcal{I}}}(S) \subseteq V^2$.

Remark 3.1. *By construction, we immediately obtain the following:*

$$G_{\mathcal{I}}^* \text{ contains no edge} \implies (G, \tilde{G}) \text{ is directly decidable.} \quad (3.40)$$

We now state our final graphical result, which addresses non-directly decidable pairs of ADMGs by reducing the original problem of searching for a valid vertex cover satisfying the path rank constraints over the entire induced graph $G_{\mathcal{I}}$ to the corresponding problem on the reduced graph $G_{\mathcal{I}}^*$, which is often substantially easier.

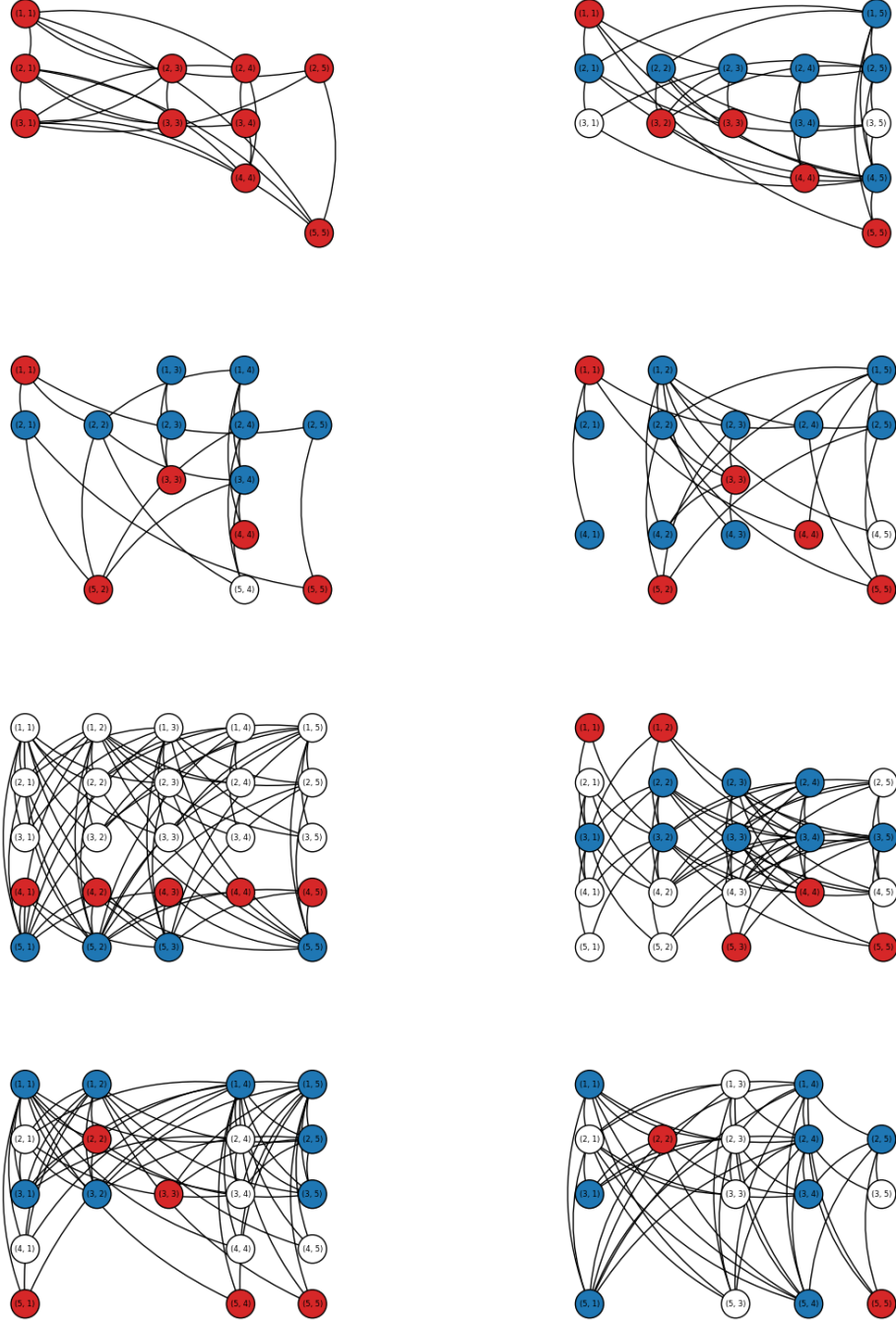


Figure 11: Induced graphs with size $|V|^2 = 5^2$: the red vertices represent the unselectable set S while the blue vertices are their neighbors in G_T^X . The first four induced graphs correspond to directly decidable pairs, whereas the last four correspond to non-directly decidable pairs.

Table 4: The naive and improved approach.

Naive approach (Theorem 3.7)	Improved approach (Theorems 3.18 and 3.21 and corollary 3.19)
1. Enumerate all vertex covers 2. Check path rank constraints 3. Return true if a valid cover exists	1. Compute relevant and unselectable sets 2. Check necessary conditions 3. Decide directly if possible 4. Construct the reduced graph 5. Enumerate inclusive minimal covers 6. Return true if a valid cover exists

Theorem 3.21. *Let $(G, \tilde{G})$ be a non-directly decidable pair of ADMGs, then $\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}$ if and only if there exists a vertex cover Q of $G_{\mathcal{I}}^*$ so that*

$$r_G(J_i, \text{pa}_{\tilde{G}}(i)) = r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i) \quad \text{for all } i \in V. \quad (3.41)$$

Here we define $J_i := \{j \in V : (i, j) \in \text{ne}_{G_{\mathcal{I}}^X}(S) \cup Q\}$ and $X, S \subseteq V^2$ are defined as in (3.35) and (3.36).

Proof. If $(G, \tilde{G})$ is an non-directly decidable pair of ADMGs, then $G_{\mathcal{I}}^*$ has uncovered edges. For the necessary direction, let Q' be a vertex cover of $G_{\mathcal{I}}^*$ satisfying the path rank constraints. We see $Q := Q' \cap (X \setminus \text{cl}_{G_{\mathcal{I}}}(S))$ is then a vertex cover of $G_{\mathcal{I}}^*$. To show that this specific Q also satisfies the path rank constraints, i.e., $r_G(J_i, \text{pa}_{\tilde{G}}(i)) = r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup i)$ for all $i \in V$ with $J_i := \{j : (i, j) \in \text{ne}_{G_{\mathcal{I}}^X}(S) \cup Q\}$, we first notice that Q' must contain the neighbors of the unselectable set, so that we have $Q' \supseteq Q \cup \text{ne}_{G_{\mathcal{I}}^X}(S)$. By Lemma 3.14, we know that subsets of a consistent set are all consistent, thus the path rank constraints hold for $Q \cup \text{ne}_{G_{\mathcal{I}}^X}(S)$. For the sufficient direction, we only need to show that $Q \cup \text{ne}_{G_{\mathcal{I}}^X}(S)$ is a vertex cover of $G_{\mathcal{I}}^X$. As $(G, \tilde{G})$ is an indirectly decidable pair of ADMGs, we know there is no edge in $G_{\mathcal{I}}^X$ with both endpoints in S . This implies that any edge in $G_{\mathcal{I}}^X$ either has an endpoint in $\text{ne}_{G_{\mathcal{I}}^X}(S)$, or has both endpoints in $X \setminus \text{cl}_{G_{\mathcal{I}}^X}(S)$. The first case is ensured by taking $\text{ne}_{G_{\mathcal{I}}^X}(S)$ into our vertex cover and the second case is ensured by taking Q into our vertex cover. $\square$

In practice, we never need to check all vertex covers. More specifically, for deciding model equivalence, it is always sufficient to check the path rank constraints only on all inclusive minimal vertex covers. A vertex cover $Q \subseteq V$ of an undirected graph G is called **inclusive minimal** if there is no strict subset $Q' \subsetneq Q$ that is also a vertex cover of G . This follows naturally from Corollary 3.15: if a vertex cover Q fails to satisfy the path rank constraints, then no larger vertex cover can satisfy them.

All the previous results allow us to simplify the algorithm, as summarized in Table 4 and illustrated in Figure 12. A practical implementation of the improved approach is presented in Algorithm 1. Code is available here.

From the analysis of our improved algorithm Algorithm 1, we observe that the only potential source of computational complexity arises from the final step, where all inclusive minimal vertex covers of the reduced graph are enumerated when none of the previous efficient checks succeeds. We now focus on addressing this challenge. For simplicity, we will henceforth refer to an inclusive minimal vertex cover simply as a **minimal vertex cover**. Note that this terminology is distinct from “minimum” vertex covers, which refer to vertex covers with the smallest cardinality.

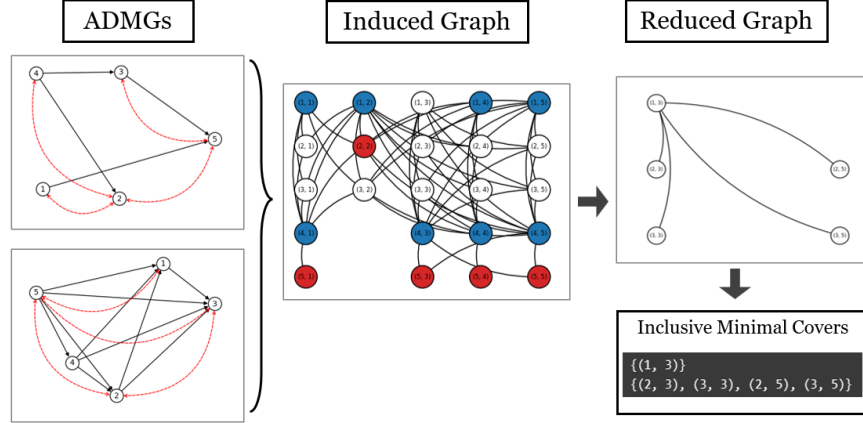


Figure 12: The improved approach. Singletons in the reduced graph are removed for clarity as they are irrelevant for vertex covers.

Algorithm 1: Model inclusion check

Input: Two ADMGs G and $\tilde{G}$

Output: Whether $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$

```

1 Construct the induced graph  $G_I(G, \tilde{G})$ ;
2 #Determine special vertices as in Definition 3.6
3 Compute the relevant set  $X$  and the unselectable set  $S$ ;
4 Project  $G_I$  onto  $G_I^X$ ;
5 if there exists an edge in  $G_I^X$  with both endpoints in  $S$  then
6   #Fast return
7   return false;
8 Compute  $\text{ne}_{G_I^X}(S)$ ;
9 foreach  $i \in V$  do
10   Let  $J_i = \{j : (i, j) \in \text{ne}_{G_I^X}(S)\}$ ;
11   if  $r_G(J_i, \text{pa}_{\tilde{G}}(i)) \neq r_G(J_i, \text{pa}_{\tilde{G}}(i) \cup \{i\})$  then
12     #Fast return based on Corollary 3.19
13     return false;
14 if  $\text{ne}_{G_I^X}(S)$  is a vertex cover of  $G_I^X$  then
15   #Fast return based on Theorem 3.18
16   return true;
17 #If still no return
18 Construct the reduced graph  $G_I^*$ ;
19 #We need to enumerate all inclusive minimal vertex covers
20 foreach inclusive minimal vertex cover  $Q$  of  $G_I^*$  do
21   foreach  $i \in V$  do
22     Let  $J_i^* = \{j : (i, j) \in Q\} \cup J_i$ ;
23     if  $r_G(J_i^*, \text{pa}_{\tilde{G}}(i)) \neq r_G(J_i^*, \text{pa}_{\tilde{G}}(i) \cup \{i\})$  then
24       break;
25   return true;
26 return false;
```

A minimal vertex cover enumeration algorithm cannot generally be polynomial-time in the input size only, because a graph with $|V|$ vertices can have exponentially many minimal vertex covers. For example, a graph consisting of $|V|/2$ disjoint edges has $2^{|V|/2}$ minimal vertex covers, since for each edge we can independently choose one endpoint. Therefore the best target is **polynomial delay** [Johnson et al., 1988]:

Definition 3.8 (Polynomial delay). *An enumeration algorithm is said to have polynomial delay if the time required to generate*

- *the first output and*
- *the delay between any two consecutive outputs*

are bounded by a polynomial in the input size.

The key point is that the total running time may still be exponential, because the number of outputs itself may be exponential. Specifically, it has been shown that the number of minimal vertex covers is bounded by $3^{|V|/3}$ [Moon and Moser, 1965]. However, a polynomial-delay algorithm guarantees that the time required to generate each minimal vertex cover is polynomial in the size of the reduced induced graph $G_{\mathcal{I}}^*$. Therefore, when $G_{\mathcal{I}}^*$ has only a moderate number of minimal vertex covers, the enumeration remains computationally feasible. This type of complexity guarantee is often referred to as **output-sensitive** enumeration.

The main question is then whether such polynomial-delay enumeration can be achieved. This topic has been studied in view of enumerating **maximal cliques** and **maximal independent sets** [Tsukiyama et al., 1977, Johnson et al., 1988, Kobayashi et al., 2024]. A maximal independent set is an independent set that is not a subset of any other independent set. A maximal clique is a set of vertices that induces a complete subgraph, and that is not a subset of the vertices of any larger complete subgraph. These concepts are related in the following way:

$$\begin{aligned} S \text{ is a maximal independent set in } G \\ \iff S \text{ is a maximal clique in } G^c \\ \iff S^c \text{ is a minimal vertex cover of } G, \end{aligned}$$

where S^c denotes the complement of S in G and G^c denotes the complementary graph of G .

Importantly, Tsukiyama et al. [1977] showed that all maximal independent sets can be enumerated with $\mathcal{O}(|V| + |E|)$ delay, where $|E|$ denotes the number of edges. This gives positive answer to our question and we conclude that our Algorithm 1 has worst case polynomial delay processing time in the size of minimal vertex covers:

Proposition 3.22. *Algorithm 1 has $\mathcal{O}(|V|^7)$ delay if the $\mathcal{O}(|V|^4)$ deciding criteria fail. The worst case running time is bounded by $\mathcal{O}(|V|^7 \cdot N(G_{\mathcal{I}}^*))$, where $N(G_{\mathcal{I}}^*)$ denotes the number of minimal vertex covers in $G_{\mathcal{I}}^*$.*

Proof. The proposed algorithm Algorithm 1 first run the fast deciding criteria in $\mathcal{O}(|V|^4)$, if it fails to decide, we need to enumerate all minimal vertex covers. This enumeration step has quadratic delay in the size of the vertices. This is simply $\mathcal{O}(|V|^{2 \times 2})$. Moreover, for each of such covers, we need to run the maximum flow algorithm in $\mathcal{O}(|V|^3)$. The overall time is bounded by $N(G_{\mathcal{I}}^*)$. $\square$

Proposition 3.22 provides a rather loose upper bound on the running time and is expected to be refined. The main purpose is to demonstrate that, within this enumeration paradigm, polynomial delay is the strongest achievable guarantee in general. We argue that obtaining an

overall polynomial running time would require incorporating additional structural constraints into the algorithm.

Nevertheless, Algorithm 1 exhibits stable performance on moderate-sized graphs with up to 20 vertices (see Figure 13). In particular, the proposed approach consistently decides model inclusion within half a second for graphs with fewer than 15 vertices, making it practically applicable for potential discovery methods.

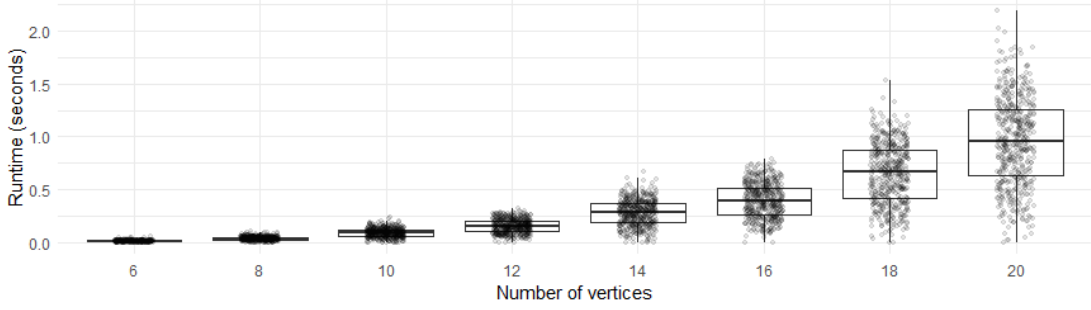


Figure 13: Runtime performance of Algorithm 1 on graphs with $|V| \leq 20$. For each fixed value of $|V|$, 500 graph instances are sampled. Boxplots show the runtime distribution, and individual points represent single runs.

3.3 Edge Removal

A graphical characterization of model equivalence has now been found. However, the resulting criterion is difficult to verify in practice. We therefore consider the simpler and more local problem of edge removal: under what conditions can an edge be removed without changing the equivalence class? Fortunately, in this setting, we obtain a simple local graphical criterion.

Remark: The development and proof of Theorem 3.23 were mainly due to Daniele Tramontano.

Definition 3.9 (Removable ancestors). *Given an ADMG G . The set of removable ancestors of a node $v \in V$ is defined as*

$$\begin{aligned} R_v &:= \{u \in \text{an}(v) : \text{Sib}(u) \setminus \text{Sib}(v) \neq \emptyset\} \\ &= \text{Sib}(V \setminus \text{Sib}(v)) \cap \text{an}(v). \end{aligned} \quad (3.42)$$

Theorem 3.23. *Let $G = (V, E_{\rightarrow}, E_{\leftrightarrow})$ be an ADMG containing an edge $e = u \rightarrow v \in E_{\rightarrow}$. Let $G^e := (V, E_{\rightarrow} \setminus \{e\}, E_{\leftrightarrow})$. Let R_v be the removable ancestors of v in G . Then G and G^e are model equivalent if and only if*

$$r_G(R_v, \text{pa}_G(v) \setminus u) = r_G(R_v, \text{pa}_G(v)). \quad (3.43)$$

Proof. Since G^e is a subgraph of G , we always have $\mathcal{M}(G^e) \subseteq \overline{\mathcal{M}(G)}$ by considering the trivial parametrization $\lambda_e = \lambda_{uv} = 0$ in G and $\lambda_{ij} = \tilde{\lambda}_{ij}$ otherwise. For the other direction, let $X = (I - \Lambda)^{-T} \varepsilon \in \mathcal{M}(G)$. Then $X \in \overline{\mathcal{M}(G^e)}$ if and only if $X \stackrel{d}{=} (I - \tilde{\Lambda})^{-T} \tilde{\varepsilon}$ for a generic $\tilde{\Lambda} \in \mathbb{R}^{G_D^e}$ and $\tilde{\varepsilon} \in \mathcal{M}(G_B^e)$. This implies

$$\tilde{\varepsilon} \stackrel{d}{=} (I - \tilde{\Lambda})^T (I - \Lambda)^{-T} \varepsilon = \underbrace{(I - \tilde{\Lambda})^T B_{\Lambda}}_A \varepsilon. \quad (3.44)$$

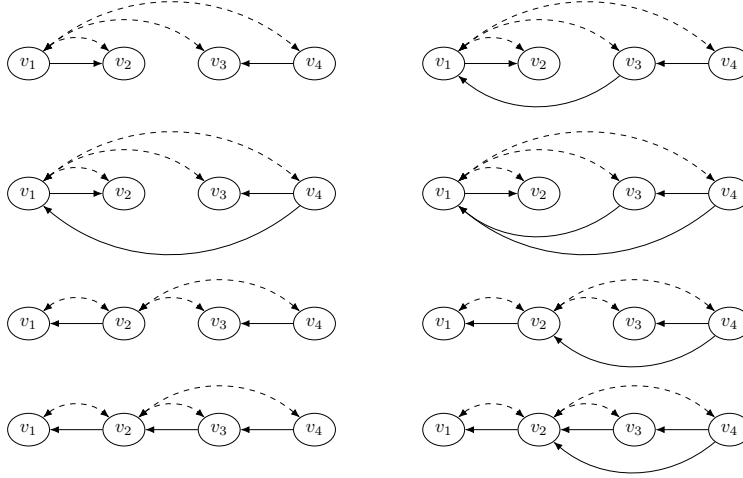


Figure 14: An equivalence class of ADMGs with four vertices. From upper left, row by row: $G_1, G_2, \dots, G_8$. All ADMGs in this class can be derived from G_4 or G_8 by removing directed edges.

Since G is acyclic, there exists a permutation on the vertices $\sigma(V(G))$ so that the G_D obeys a topological ordering. It is clear that $\sigma(G_D^e)$ also follows a topological ordering. We consider the equivalent problem $\mathcal{M}(\sigma(G)) \subseteq \overline{\mathcal{M}(\sigma(G^e))}$. This allows a simpler representation of A , namely, we have

$$a_{uv} = b_{uv} - \sum_{w \in \text{pa}(u)^e \cap \text{de}(v)} \tilde{\lambda}_{wu} b_{wv},$$

where $\text{pa}(v)^e$ is the parent set of v in G^e . Since G and G^e have the same bidirected part, we can conclude that a matrix A as in Equation (3.44) can exist if and only if, for every $v \in V$, the following system has a solution:

$$[(B_\Lambda)_{\text{pa}(v)^e, R_v}]^T \cdot \tilde{\lambda}_{\text{pa}(v)^e, v} = [(B_\Lambda)_{v, R_v}]^T. \quad (3.45)$$

Let $e = u_0 \rightarrow v_0$. For $v \neq v_0$, we have $\text{pa}(v)^e = \text{pa}(v)$. Hence, the system in Equation (3.45) has always a solution given by $\tilde{\lambda}_{\text{pa}(v)^e, v} = \lambda_{\text{pa}(v), v}$. For $v = v_0$, we have $\text{pa}(v_0)^e = \text{pa}(v_0) \setminus \{u_0\}$. The system in Equation (3.45) has a solution if and only if $\text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\}, R_{v_0}}) = \text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}, R_{v_0}})$. Let $\Pi = (\pi_0, \dots, \pi_k)$ be a system of paths without intersection from $I \subseteq R_{v_0}$ to $J \subseteq \text{pa}(v_0)$. If $u_0 \in \Pi$, let π_0 be the path that ends at u_0 , and π_0^* be the path obtained by concatenating π_0 with the edge $u_0 \rightarrow v_0$. By contradiction, $\Pi^* = (\pi_0^*, \dots, \pi_k)$ is a system of non-intersection paths from I to $J \setminus \{u_0\} \cup \{v_0\} \subseteq \text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}$. This proves that $\text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}, R_{v_0}}) = \text{rank}((B_\Lambda)_{\text{pa}(v_0), R_{v_0}})$. We know that $\text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\}, R_{v_0}}) \geq \text{rank}((B_\Lambda)_{\text{pa}(v_0), R_{v_0}}) - 1$, and we can write

$$\begin{aligned} \text{rank}((B_\Lambda)_{\text{pa}(v_0), R_{v_0}}) - 1 &\leq \text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\}, R_{v_0}}) \\ &\leq \text{rank}((B_\Lambda)_{\text{pa}(v_0) \setminus \{u_0\} \cup \{v_0\}, R_{v_0}}) = \text{rank}((B_\Lambda)_{\text{pa}(v_0), R_{v_0}}). \end{aligned} \quad (3.46)$$

We conclude the proof by noticing that the second inequality becomes equal if and only if the first inequality is strict, and that implies $r_G(R_{v_0}, \text{pa}(v_0) \setminus u_0) = r_G(R_{v_0}, \text{pa}(v_0))$. $\square$

Example 3.7. Consider G_1 and G_5 from Figure 3. Let $G := G_5$ and $G^e = G_1$. Let $e = v_1 \rightarrow v_3$ be the removed edge. By definition, $R_{v_3} = \text{Sib}(V \setminus \text{Sib}(v_3)) \cap \text{an}(v_3) = \{v_1\}$. Further, we have

$$r_G(v_1, \text{pa}_G(v_3) \setminus v_1) = r_G(v_1, \text{pa}_G(v_3)) = 1.$$

So we conclude that removing $v_1 \rightarrow v_3$ from G keeps it in the same equivalence class. Using the same approach, we can also verify that $\mathcal{M}(G_5) = \mathcal{M}(G_2)$ by removing $v_2 \rightarrow v_3$, $\mathcal{M}(G_4) = \mathcal{M}(G_2)$ by removing $v_3 \rightarrow v_2$, and $\mathcal{M}(G_4) = \mathcal{M}(G_3)$ by removing $v_1 \rightarrow v_2$. As illustrated in Figure 15, the whole equivalence class of the IV graph in Figure 3 can be derived from any ADMG in this class by sequentially adding or removing edges.

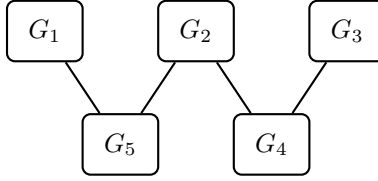


Figure 15: The relation of $G_1 \dots, G_5$ from Figure 3. An undirected edge between two ADMGs represents that one could be derived from another by adding or removing a directed edge.

Theorem 3.23 provides a graphical check if a directed edge is removable when considering model equivalence. While verifying Equation (3.43) directly could be computationally challenging, Tramontano et al. [2026] proposed algorithms that can verify such a condition in polynomial time in the size of the graph via a maximum flow reformulation. In fact, Equation (3.43) is related to the concept of generic identifiability (Tramontano et al. [2026, §2]). Essentially, Theorem 3.23 states that a directed edge $u \rightarrow v$ can be removed without changing the equivalence class if and only if the parameter λ_{uv} is not generically identifiable.

This result provides an important theoretical step toward the development of structure learning algorithms. In particular, if one could prove that, after fixing the confounding structure, every graph in an equivalence class can be transformed into any other graph in the same class through a sequence of edge additions, removals and reversals, then one could efficiently learn the entire equivalence class (with a fixed confounding structure) by developing a traversal algorithm using DFS/BFS. The hypothesis that every pair of equivalent graphs can be transformed into one another in this way is known as the “Meek conjecture” (or traversal conjecture) [Meek, 1997]. Figure 15 provides such an example. This conjecture has recently been proven for the lv-LiNGM setting, in which latent confounding is assumed to affect the distribution linearly [Dai et al., 2026].

We want to remark that the traversal conjecture is true for $|V| = 3$. However, the conjecture is in general **false** in our general confounded setting when fixing a confounding structure. For $|V| = 4$, we tested all 1165 non-isomorphic, non-trivial equivalence classes and found 7 counterexamples (code). One of such counterexample is provided in Figure 16. We want to remark that this conjecture is also false when allowing cycles. We will briefly discuss the conjecture for the cyclic case at the end of Section 3.6.

3.4 Edge Reversal

We now focus on another local characterization problem: when can a directed edge be reversed while remaining in the equivalence class? Notice that in Section 3.3, we found a simple graphical characterization for directed edge removal and addition that requires computing only two path

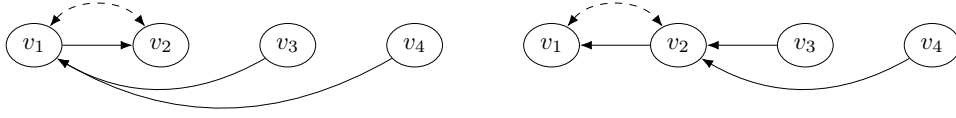


Figure 16: A counterexample to the traversal conjecture. This equivalence class consists of exactly two ADMGs that cannot be transformed into each other by a sequence of edge additions, deletions, and reversals while remaining within the equivalence class.

ranks. Ideally, we would like to adopt this edge deletion and addition operation to obtain a characterization for edge reversal, provided that we can prove that a directed edge can only be reversed if it can also be removed.

For simplicity, let $G^{\rightarrow}$ be an ADMG and let $e = u \rightarrow v \in G^{\rightarrow}$ be a directed edge in it. We define $G^{\leftarrow}$ to the ADMG with edge e being reversed and G^e be the ADMG with edge e being removed. The main result is the following.

Theorem 3.24. *Assume $u \leftrightarrow v \in G_B^{\rightarrow}$. Then the following holds.*

$$\overline{\mathcal{M}(G^{\rightarrow})} = \overline{\mathcal{M}(G^{\leftarrow})} \iff \overline{\mathcal{M}(G^{\rightarrow})} = \overline{\mathcal{M}(G^e)} = \overline{\mathcal{M}(G^{\leftarrow})}. \quad (3.47)$$

Proof. The reversed direction is clear. To prove the “ $\implies$ ” direction, we need to assume $\overline{\mathcal{M}(G^{\rightarrow})} = \overline{\mathcal{M}(G^{\leftarrow})}$, and show that $\overline{\mathcal{M}(G^{\rightarrow})} = \overline{\mathcal{M}(G^e)}$. The second part $\overline{\mathcal{M}(G^{\leftarrow})} = \overline{\mathcal{M}(G^e)}$ can be proved similarly. As $\overline{\mathcal{M}(G^{\rightarrow})} \supseteq \overline{\mathcal{M}(G^e)}$ is clear by model parametrization, it remains to show $\overline{\mathcal{M}(G^{\rightarrow})} \subseteq \overline{\mathcal{M}(G^e)}$. To show this, by Theorem 3.7, we only need to find a vertex cover $Q \subseteq V^2$ of the induced graph $G_{\mathcal{I}}(G_B^{\rightarrow}, G_B^e)$ such that for all $i \in V$:

$$r_{G^{\rightarrow}}(Q_i, \text{pa}_{G^e}(i)) = r_{G^{\rightarrow}}(Q_i, \text{pa}_{G^e}(i) \cup i) \quad (3.48)$$

with $Q_i := \{j \in V : (i, j) \in Q\}$. As $G^{\rightarrow}, G^{\leftarrow}, G^e$ have the same bidirected part, the induced graphs between them are always the same, and we simply write $G_{\mathcal{I}}$ for the common induced graph. By assumption that $\overline{\mathcal{M}(G^{\rightarrow})} \subseteq \overline{\mathcal{M}(G^{\leftarrow})}$, again using Theorem 3.7, we know there exists a vertex cover $Q' \subseteq V^2$ of the induced graph $G_{\mathcal{I}}$ such that for all $i \in V$:

$$r_{G^{\rightarrow}}(Q'_i, \text{pa}_{G^{\leftarrow}}(i)) = r_{G^{\rightarrow}}(Q'_i, \text{pa}_{G^{\leftarrow}}(i) \cup i) \quad (3.49)$$

with $Q'_i := \{j \in V : (i, j) \in Q'\}$. If we can show that this particular vertex cover Q' satisfies all the path rank constraints in Equation (3.48), then we are done. To prove this, notice that for all $i \in V \setminus u$, we have $\text{pa}_{G^{\leftarrow}}(i) = \text{pa}_{G^e}(i)$, so we only need to check the path rank condition for the case $i = u$. For $i = u$, we have $\text{pa}_{G^{\leftarrow}}(u) = \text{pa}_{G^e}(u) \cup v = \text{pa}_{G^{\rightarrow}}(u) \cup v$, and Equation (3.49) becomes

$$r_{G^{\rightarrow}}(Q'_u, \text{pa}_{G^{\rightarrow}}(u) \cup v) = r_{G^{\rightarrow}}(Q'_u, \text{pa}_{G^{\rightarrow}}(u) \cup u \cup v). \quad (3.50)$$

What we want to prove is $r_{G^{\rightarrow}}(Q'_u, \text{pa}_{G^e}(u)) = r_{G^{\rightarrow}}(Q'_u, \text{pa}_{G^e}(u) \cup u)$. As $\text{pa}_{G^e}(u) = \text{pa}_{G^{\rightarrow}}(u)$, this equation simplifies to

$$r_{G^{\rightarrow}}(Q'_u, \text{pa}_{G^{\rightarrow}}(u)) = r_{G^{\rightarrow}}(Q'_u, \text{pa}_{G^{\rightarrow}}(u) \cup u). \quad (3.51)$$

The only remaining task is to show that Equation (3.50) implies Equation (3.51). Notice that Equation (3.51) holds if and only if $u \notin Q'_u$. To see this, assume $r_{G^{\rightarrow}}(Q'_u, \text{pa}_{G^{\rightarrow}}(u)) \neq r_{G^{\rightarrow}}(Q'_u, \text{pa}_{G^{\rightarrow}}(u) \cup u)$, this means $r_{G^{\rightarrow}}(Q'_u \cup v, \text{pa}_{G^{\rightarrow}}(u)) + 1 = r_{G^{\rightarrow}}(Q'_u \cup v, \text{pa}_{G^{\rightarrow}}(u) \cup u)$, and

we conclude that u must be on an activated path which does not intersect with $\text{pa}_{G \rightarrow}(u)$, since all non-trivial paths from Q'_u to u must either go through $\text{pa}_{G \rightarrow}(u)$, or are themselves in $\text{pa}_{G \rightarrow}(u)$, the only possibility is a trivial path.

Until now, we showed that the original claim can be reformulated as the following claim, namely, we need to prove that, if $\mathcal{M}(G \rightarrow) \subseteq \overline{\mathcal{M}(G \leftarrow)}$, then there exists a vertex cover $Q \subseteq V^2 \setminus (u, u)$ for the induced graph $G_{\mathcal{I}}(G_B^{\rightarrow}, G_B^{\leftarrow})$ so that for all $i \in V$ we have:

$$r_{G \rightarrow}(Q_i, \text{pa}_{G \leftarrow}(i)) = r_{G \rightarrow}(Q_i, \text{pa}_{G \leftarrow}(i) \cup i) \quad (3.52)$$

with $Q_i := \{j \in V : (i, j) \in Q\}$. The main point is to show that the vertex cover Q can be chosen in the way that $(u, u) \notin Q$. By permuting $G \rightarrow$ to a topological order, we can show that (i, i) with $i \in V \setminus u$ can never be selected in Q using Lemma 3.17 as we can verify that $A_{ii} = 1$. Moreover, (v, u) can also never be selected in Q if we restrict ourselves to the acyclic setting. This is because by Definition 3.6, we can show that $v \in \text{an}_{G \rightarrow}(u) \setminus \text{an}_{G \rightarrow}(\text{pa}_{G \leftarrow}(u))$ and thus (v, u) is not selectable. To see $u \in \text{an}_{G \rightarrow}(v) \setminus \text{an}_{G \rightarrow}(\text{pa}_{G \leftarrow}(v))$ holds, we first notice that $\text{pa}_{G \leftarrow}(v) = \text{pa}_{G \rightarrow}(v) \setminus u$. Since it is clear that $u \in \text{pa}_{G \rightarrow}(v)$, it remains to show that $u \notin \text{an}_{G \rightarrow}(\text{pa}_{G \rightarrow}(v) \setminus u)$, and this is the same of ensuring no alternative path from u to v . Since we assumed that the edge $u \rightarrow v$ can be reversed without creating cycles, we can conclude that such an alternative path cannot exist and $u \notin \text{an}_{G \rightarrow}(\text{pa}_{G \rightarrow}(v) \setminus u)$. Let us denote $N = \{(i, i) : i \in V \setminus u\} \cup \{(v, u)\}$. We know that for all vertex cover $Q \subseteq V^2$ satisfying the path rank constraints we have $Q \cap N = \emptyset$. Let $Q^+ := V^2 \setminus (N \cup \{(u, u)\})$. The next step is to show that

$$r_{G \rightarrow}(Q_i^+, \text{pa}_{G \leftarrow}(i)) = r_{G \rightarrow}(Q_i^+, \text{pa}_{G \leftarrow}(i) \cup i) \quad (3.53)$$

with $Q_i^+ := \{j \in V : (i, j) \in Q^+\}$. To prove this, we need to show that for all $i \in V$, the linear system $L_i := \{A_{ij} : j \in Q_i^+\}$ with A as defined in Theorem 3.1 is generically solvable in $\tilde{\Lambda}$. Notice that A is lower triangular with ones on the diagonal when ignoring the row u . In row u , we have $A_{uk} = 0$ for $k > v$. For $i \notin \{u, v\}$, the linear system L_i can be solved ladderwise with general solution

$$\begin{aligned} \tilde{\lambda}_{i-1,i} &= b_{i,i-1} \\ \tilde{\lambda}_{i-2,i} &= -b_{i-1,i-2}\tilde{\lambda}_{i-1,i} + b_{i,i-2} \\ &\dots \\ \tilde{\lambda}_{1i} &= -\sum_{1 < r < i} b_{r1}\tilde{\lambda}_{ri} + b_{i1}, \end{aligned}$$

where $b_{ij} = (I - \Lambda)_{ij}^{-1}$. For $i = u$, a canonical solution for L_u is given by

$$\begin{aligned} \tilde{\lambda}_{vu} &= 0 \\ \tilde{\lambda}_{u-1,u} &= -b_{v,u-1}\tilde{\lambda}_{vu} + b_{u,u-1} \\ &\dots \\ \tilde{\lambda}_{1,u} &= -b_{v,1}\tilde{\lambda}_{vu} - \sum_{1 < r < u} b_{r1}\tilde{\lambda}_{ru} + b_{u1}. \end{aligned}$$

For $i = v$, a canonical solution for L_v is given by

$$\begin{aligned}\tilde{\lambda}_{v-1,v} &= b_{v,v-1} \\ \tilde{\lambda}_{v-2,v} &= -b_{v-1,v-2}\tilde{\lambda}_{v-1,v} + b_{v,v-2} \\ &\dots \\ \tilde{\lambda}_{1v} &= - \sum_{\substack{1 \leq r \leq v \\ r \neq u}} b_{r1}\tilde{\lambda}_{rv} + b_{v1}.\end{aligned}$$

So far, we showed that Q^+ is indeed valid considering all path rank constraints. It only remains to show that Q^+ is a valid vertex cover. To prove this, we need to show that there is no edge in $G_{\mathcal{I}}$ with both endpoints in $N \cup \{(u, u)\}$. Assume that there exists an edge in $G_{\mathcal{I}}$ with both endpoints in N , then no valid vertex cover satisfying all path rank constraints can exist, which is a contradiction to $\mathcal{M}(G^{\rightarrow}) \subseteq \overline{\mathcal{M}(G^{\leftarrow})}$. It follows that the only possible trouble is an edge connecting (u, u) and a vertex in N . Notice that an edge in $G_{\mathcal{I}}$ connecting (u, u) and (i, i) with $i \in V \setminus u$ would mean that $u \leftrightarrow i \in G$ and $u \leftrightarrow i \notin G$, which is not possible. So we can finally conclude that the only edge in $G_{\mathcal{I}}$ which is not always covered by Q^+ is the edge between (u, u) and (v, u) . However, we already ruled out this case by assuming $u \leftrightarrow v \in G$. $\square$

We remark that Theorem 3.24 uses the assumption that there is a bidirected edge connecting the endpoints of the reversed directed edge. Can this assumption be dropped? We run computer experiments by sampling cases in which an edge can be reversed and then examine whether there is a bidirected edge connecting them. After testing 10,000 reversible cases with $|V| = 5$, we did not find a single counterexample (code). We therefore conjecture that this may always be true.

Conjecture 1. *If a directed edge $u \rightarrow v$ can be reversed while remaining model equivalent, then $u \leftrightarrow v \in G$.*

The intuition behind Conjecture 1 is that, if there is only one causal direction in the causal structure that we cannot determine, namely, the one between u and v , then the corresponding non-Gaussian noise terms ε_u and ε_v should generically be dependent. The major obstacle to proving the above conjecture is that the global confounding structure can be arbitrary, making a local analysis of the dependence between ε_u and ε_v challenging.

Proposition 3.25. *If Conjecture 1 is true, then*

$$\overline{\mathcal{M}(G^{\rightarrow})} = \overline{\mathcal{M}(G^{\leftarrow})} \iff \overline{\mathcal{M}(G^{\rightarrow})} = \overline{\mathcal{M}(G^e)} = \overline{\mathcal{M}(G^{\leftarrow})}, \quad (3.54)$$

i.e., a directed edge can be reversed if and only if it can be removed and added back backwards using the criterion in Theorem 3.23.

Proof. This is clear by verifying the assumption used in the proof of Theorem 3.24. $\square$

3.5 Constraints on the Confounding Structure

We now turn our attention to the confounding structure and ask the following question: How can the bidirected part vary within an equivalence class? By the nonparametric nature of our confounding model, we expect that model equivalence between two DMGs imposes necessary constraints on their bidirected components. Such constraints are useful, as they can significantly reduce the space of candidate equivalent models. Non-Gaussian models admit stronger identifiability results, since non-Gaussian distributions contain higher-order moment information

beyond covariance structure. Cumulant-based method is the state-of-the-art tool for studying such embedded information [cf. Robeva and Seby, 2021, Cai et al., 2023, Tramontano et al., 2025, Drton et al., 2025a]. In the following, we establish a general result based on connectedness constraints. Using this tool, we can show as a special case that if two DMGs are model equivalent, then they must have the same number of bidirected edges.

Definition 3.10 (Cumulant tensor). *The k -th cumulant tensor of a random vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_p)$ is the k -way tensor in $(\mathbb{R}^p)^k$ with*

$$\mathcal{C}^{(k)}(\varepsilon)_{i_1, \dots, i_k} := \sum_{(A_1, \dots, A_L)} (-1)^{L-1} (L-1)! \cdot \mathbb{E} \left[\prod_{j \in A_1} \varepsilon_j \right] \cdots \mathbb{E} \left[\prod_{j \in A_L} \varepsilon_j \right], \quad (3.55)$$

where the sum is taken over all partitions $(A_1, \dots, A_L)$ of the **multiset** $\{i_1, \dots, i_k\}$ with arbitrary $L \in \{1, \dots, k\}$.

We always center ε so that $\mathcal{C}^{(1)}(\varepsilon) = \mathbb{E}[\varepsilon] = 0$. For $k = 2$, $\mathcal{C}^{(2)}(\varepsilon)$ is the covariance matrix. Moreover, $\mathcal{C}^{(3)}(\varepsilon)_{ijk} = \mathbb{E}[\varepsilon_i \varepsilon_j \varepsilon_k]$ and

$$\mathcal{C}^{(4)}(\varepsilon)_{ijkl} = \mathbb{E}[\varepsilon_i \varepsilon_j \varepsilon_k \varepsilon_l] - \mathbb{E}[\varepsilon_i \varepsilon_j] \mathbb{E}[\varepsilon_k \varepsilon_l] - \mathbb{E}[\varepsilon_i \varepsilon_k] \mathbb{E}[\varepsilon_j \varepsilon_l] - \mathbb{E}[\varepsilon_i \varepsilon_l] \mathbb{E}[\varepsilon_j \varepsilon_k]. \quad (3.56)$$

Cumulant tensors are symmetric, i.e.,

$$\mathcal{C}^{(k)}(\varepsilon)_{i_1, \dots, i_k} = \mathcal{C}^{(k)}(\varepsilon)_{\sigma(i_1), \dots, \sigma(i_k)} \quad \forall \sigma \in S_k, \quad (3.57)$$

where S_k is the symmetric group on $\{1, \dots, k\}$, and σ is a permutation on it. We write $\text{Sym}_k(p)$ for the subspace of symmetric tensors in $(\mathbb{R}^p)^k$.

Lemma 3.26 (Jutten and Comon [2010]). *Let $\varepsilon = (\varepsilon_1, \dots, \varepsilon_p)$ be any p -variate random vector, and $A \in \mathbb{R}^{s \times p}$ for any $s \in \mathbb{N}$, then*

$$\mathcal{C}^{(k)}(A\varepsilon)_{i_1, \dots, i_k} = \sum_{1 \leq j_1, \dots, j_k \leq p} \mathcal{C}^{(k)}(\varepsilon)_{j_1, \dots, j_k} a_{j_1, i_1} \cdots a_{j_k, i_k} \quad (3.58)$$

$$= (\mathcal{C}^{(k)}(\varepsilon) \bullet_k A)_{i_1, \dots, i_k}, \quad (3.59)$$

where $\bullet_k$ denotes the Tucker product.

Higher moment information is embedded in the cumulant tensor space.

Definition 3.11 (Cumulant tensor space). *Let*

$$\begin{aligned} \mathcal{T}^{(k)}(G_B) := \{ \mathcal{C}^{(k)} \in \text{Sym}_k(\mathbb{R}^p) : \mathcal{C}_{i_1, \dots, i_k}^{(k)} = 0 \\ \text{if } \{i_1, \dots, i_k\} \text{ is not connected in } G_B \}, \end{aligned} \quad (3.60)$$

and let

$$\mathcal{T}^{(k)}(G) := \{ \mathcal{C}^{(k)} \bullet_k B_\Lambda \mid \mathcal{C}^{(k)} \in \mathcal{T}^{(k)}(G_B), \Lambda \in \mathbb{R}^{G_D} \}. \quad (3.61)$$

A set $S \subseteq V$ is connected in G_B if for each pair of vertices from S , there exists a bidirected path between them with all vertices from S . Note that $B_\Lambda = (I - \Lambda)^{-T}$. We now introduce the following **connectedness constraints**.

Definition 3.12 (Connectedness constraints). *Let $G = (V, E_{\rightarrow}, E_{\leftrightarrow})$ be a DMG, and let G_B be its bidirected part. For all $i \in \mathbb{N}_{>0}$, define*

$$n_i(G_B) := |\{S \subseteq V : S \text{ connected in } G_B, |S| = i\}|. \quad (3.62)$$

We see that $n_i(G_B)$ is simply the number of connected size- i subsets of V with respect to G_B . By definition, $n_1(G_B)$ is simply the number of vertices, since each vertex is connected to itself via the trivial bidirected path. Further, $n_2(G_B)$ is the number of bidirected edges. Moreover, $n_i(G_B)$ is bounded by the binomial coefficient $\binom{p}{i}$, where $p = |V|$, and $n_i(G_B) = 0$ for $i > p$. As an example, for the ADMGs in Figure 17, $n_1(G_B) = n_1(\tilde{G}_B) = 5$, $n_2(G_B) = n_2(\tilde{G}_B) = 3$, $n_3(G_B) = n_3(\tilde{G}_B) = 1$ and $n_i(G_B) = n_i(\tilde{G}_B) = 0$ for all $i \geq 4$.

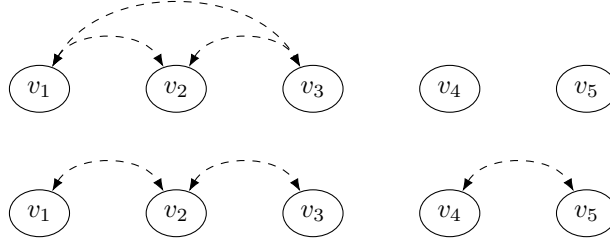


Figure 17: Bidirected parts of two DMGs: G_B (above) and $\tilde{G}_B$ (below).

The following lemma shows that the dimension of $\mathcal{T}^{(k)}(G_B)$ is solely determined by the connected constraints.

Lemma 3.27. *Let G_B be the bidirected part of an DMG G . Let $p = |V|$. Then*

$$\dim(\mathcal{T}^{(k)}(G_B)) = \sum_{i=1}^{\min\{p,k\}} n_i(G_B) \cdot \binom{k-1}{i-1}. \quad (3.63)$$

Proof. Let $\mathcal{C}_J^{(k)}$ denote a k -way cumulant tensor with an index set $J = \{j_1, \dots, j_k\}$. Let $U(J)$ be the number of unique elements in a multiset J . Let $[p] = \{1, \dots, p\}$, and let $\mathbf{1}\{\cdot\}$ be the a function which takes the value 1 if the condition is fulfilled and 0 otherwise. We can then write

$$\dim(\mathcal{T}^{(k)}(G_B)) = \sum_{\substack{j_1, \dots, j_k \in [p] \\ \mathcal{C}_J \in \text{Sym}_k(\mathbb{R}^p)}} \mathbf{1}\{\mathcal{C}_J^{(k)} \neq 0\} = \sum_{i=1}^k \sum_{\substack{j_1, \dots, j_k \in [p] \\ \mathcal{C}_J \in \text{Sym}_k(\mathbb{R}^p) \\ U(J)=i}} \mathbf{1}\{\mathcal{C}_J^{(k)} \neq 0\} \quad (3.64)$$

$$= \sum_{i=1}^k n_i(G_B) \sum_{\substack{j_1, \dots, j_k \in [i] \\ \mathcal{C}_J \in \text{Sym}_k(\mathbb{R}^i) \\ U(J)=i}} \mathbf{1} = \sum_{i=1}^{\min(p,k)} n_i(G_B) \cdot \binom{k-1}{i-1}. \quad (3.65)$$

The last equality follows from the fact that the binomial coefficient $\binom{k-1}{i-1}$ gives the number of size k multisets with their elements chosen from a size i set, where we only allow the multisets to be chosen in the way that all of the i elements are chosen at least once. Note that $k \geq \min\{p, k\} \geq i$, so the binomial coefficient is always defined. $\square$

Example 3.8. Consider $\dim(\mathcal{T}^{(3)}(G_B))$ in Figure 17. The twelve non-zero cumulant tensors are $\mathcal{C}_{1,1,1}, \dots, \mathcal{C}_{5,5,5}, \mathcal{C}_{1,1,2}, \mathcal{C}_{1,2,2}, \mathcal{C}_{1,1,3}, \mathcal{C}_{1,3,3}, \mathcal{C}_{2,2,3}, \mathcal{C}_{2,3,3}$ and $\mathcal{C}_{1,2,3}$, by ignoring all permutations.

This result corresponds to the formula

$$\dim (\mathcal{T}^{(3)}(G_B)) = \sum_{i=1}^{\min\{5,3\}} n_i(G_B) \cdot \binom{3-1}{i-1} \quad (3.66)$$

$$= 5 \cdot \binom{2}{0} + 3 \cdot \binom{2}{1} + 1 \cdot \binom{2}{2} = 12. \quad (3.67)$$

Essentially, when two DMGs are model equivalent, their corresponding cumulant tensor spaces should be of the same dimension for all $k \in \mathbb{N}_{>0}$. Our model parametrization restricts the dimensional difference of their corresponding bidirected cumulant tensor spaces. Applying the dimension formula above yields our next theorem, which provides a necessary graphical condition on the bidirected parts of two DMGs for model equivalence.

Theorem 3.28. *If two DMGs G and $\tilde{G}$ are model equivalent, then $n_i(G_B) = n_i(\tilde{G}_B)$ for all $i \in \mathbb{N}_{>0}$.*

Proof. Let us consider two DMGs G and $\tilde{G}$ which are model equivalent. Then it holds that

$$\dim (\mathcal{T}^{(k)}(G)) - \dim (\mathcal{T}^{(k)}(\tilde{G})) = 0 \quad (3.68)$$

for all $k \in \mathbb{N}$. Recall that we define the cumulant tensor space for arbitrary G to be

$$\mathcal{T}^{(k)}(G) = \{\mathcal{C} \bullet_k B_\Lambda \mid \mathcal{C} \in \mathcal{T}^{(k)}(G_B), \Lambda \in \mathbb{R}^{G_D}\},$$

where $\bullet_k$ denotes the Tucker product. This means that we can parametrize any $\hat{\mathcal{C}} \in \mathcal{T}^{(k)}(G)$ using $\mathcal{C} \in \mathcal{T}^{(k)}(G_B)$ and $\Lambda \in \mathbb{R}^{G_D}$, which implies

$$\dim (\mathcal{T}^{(k)}(G)) \leq \dim (\mathcal{T}^{(k)}(G_B)) + \dim (\mathbb{R}^{G_D}) \quad (3.69)$$

for all $k \in \mathbb{N}$. Let $c = \max \{\dim (\mathbb{R}^{G_D}), \dim (\mathbb{R}^{\tilde{G}_D})\}$ be a constant upper bound that does not depend on k . From the inequality above we obtain that

$$0 \leq \dim (\mathcal{T}^{(k)}(G)) - \dim (\mathcal{T}^{(k)}(G_B)) \leq c, \quad (3.70)$$

and similarly

$$0 \leq \dim (\mathcal{T}^{(k)}(\tilde{G})) - \dim (\mathcal{T}^{(k)}(\tilde{G}_B)) \leq c. \quad (3.71)$$

So using Equation (3.70) and Equation (3.71) we get

$$\begin{aligned} & \left| \left(\dim (\mathcal{T}^{(k)}(G)) - \dim (\mathcal{T}^{(k)}(G_B)) \right) \right. \\ & \quad \left. - \left(\dim (\mathcal{T}^{(k)}(\tilde{G})) - \dim (\mathcal{T}^{(k)}(\tilde{G}_B)) \right) \right| \leq c. \end{aligned} \quad (3.72)$$

By Equation (3.68), the inequality above simplifies to

$$|\dim (\mathcal{T}^{(k)}(G_B)) - \dim (\mathcal{T}^{(k)}(\tilde{G}_B))| \leq c, \quad (3.73)$$

whenever G and $\tilde{G}$ are model equivalent. The last inequality together with Lemma 3.27 suggest

that

$$\left| \underbrace{\sum_{i=1}^{\min(p,k)} (n_i(G_B) - n_i(\tilde{G}_B))}_{:=g(k)} \cdot \binom{k-1}{i-1} \right| \leq c. \quad (3.74)$$

We are interested in higher moments, so we will from now on look at $k \geq p$. Define $d_i := n_i(G_B) - n_i(\tilde{G}_B)$, then $g(k)$ in Equation (3.74) further simplifies to

$$g(k) = \sum_{i=2}^p \frac{d_i}{(i-1)!} \frac{(k-1)!}{(k-i)!} \quad (3.75)$$

$$= \sum_{i=2}^p \frac{d_i}{(i-1)!} (k-1)(k-2) \cdots (k-i+1). \quad (3.76)$$

Clearly $g(k)$ can be seen as a polynomial function of k since d_i and p are fixed for two given DMGs G and $\tilde{G}$. If G and $\tilde{G}$ are model equivalent, we must have

$$|g(k)| \leq c \quad \text{for all } k \geq p, k \in \mathbb{N}. \quad (3.77)$$

Consider the asymptotic behavior of $g(k)$ for k large enough. If $d_p \neq 0$, then

$$g(k) \sim \frac{d_p}{(p-1)!} k^{p-1}$$

asymptotically. Thus there exists some k large enough so that $|g(k)| > c$, which leads to a contradiction to Equation (3.77). So we conclude that $d_p = 0$. Setting d_p to zero, we then have

$$g(k) \sim \frac{d_{p-1}}{(p-2)!} k^{p-2} \quad (3.78)$$

asymptotically if $d_{p-1} \neq 0$. Similarly, we argue that $d_{p-1} = 0$ must hold as well. Continuing with the same argument, we can show that $d_i = 0$ for all $2 \leq i \leq p$, $i \in \mathbb{N}$. We conclude the proof by noticing that $d_1 = 0$ and $n_i(G_B) = n_i(\tilde{G}_B) = 0$ for $i > p$. $\square$

Note that the condition $n_i(G_B) = n_i(\tilde{G}_B)$ for all $i \in \mathbb{N}^+$ does not imply that the two bidirected parts G_B and $\tilde{G}_B$ have the same structure (also not up to symmetry); Figure 17 serves as a counterexample.

Consider a disjoint decomposition of G_B such that $V = \dot{\cup}_{i=1} D_i$, where D_i is connected in G_B . If D_i is not further contained in a larger bidirected-connected subset, then D_i is maximal and often called a **district** in G . Let

$$\mathcal{D}(G) := \{D : D \text{ is a district in } G\} \quad (3.79)$$

be the set of districts in G . Let

$$D_{\max} := \{D \in \mathcal{D}(G) : |D| \geq |D_i| \quad \forall D_i \in \mathcal{D}(G)\} \quad (3.80)$$

be the maximal district of G . A straightforward application of Theorem 3.28 yields the following results:

Corollary 3.29. *If G and $\tilde{G}$ are model equivalent DMGs, then:*

1. $|E_{\leftrightarrow}| = |\tilde{E}_{\leftrightarrow}|$,
2. $|D_{\max}| = |\tilde{D}_{\max}|$,
3. $\dim(\mathcal{T}^{(k)}(G_B)) = \dim(\mathcal{T}^{(k)}(\tilde{G}_B)) \quad \forall k \in \mathbb{N}_{>0}$.

Proof. By definition we have $n_2(G_B) = |E_{\leftrightarrow}|$, so the first equality follows. For $|D_{\max}| = |\tilde{D}_{\max}|$, note that

$$|D_{\max}| = \max \{i \in \mathbb{N}^+ : n_i(G_B) \neq 0\}.$$

The dimensional equality $\dim(\mathcal{T}^{(k)}(G_B)) = \dim(\mathcal{T}^{(k)}(\tilde{G}_B))$ for all $k \in \mathbb{N}^+$ follows directly by applying Lemma 3.27 and Theorem 3.28. $\square$

Theorem 3.28 provides a necessary condition for checking model equivalence. We aim to investigate its computational complexity. Let G and $\tilde{G}$ be two arbitrary DMGs with p nodes. Ideally, we want to derive an efficient algorithm for the non-equivalence filter

$$f(G, \tilde{G}) \mapsto \begin{cases} \text{Continue} & \text{if } n_i(G_B) = n_i(\tilde{G}_B) \quad \forall i \in [p], \\ \text{False} & \text{otherwise.} \end{cases} \quad (3.81)$$

To compute $n_i(G_B)$, we can naively go through all $\binom{p}{i}$ subsets and decide in each case whether it is connected or not. Therefore, a naive computation of f by calculating all $n_i(G_B)$ and $n_i(\tilde{G}_B)$ for $i \in [p]$ would require checking the connectedness of

$$\mathcal{O}\left(\sum_{i=2}^p \binom{p}{i}\right) \sim \mathcal{O}(2^p) \quad (3.82)$$

subsets. We may check whether a size i subset of nodes is connected in complexity $\mathcal{O}(i^2)$.

We are more interested in the case $f = \text{False}$, where we can immediately conclude non-equivalence. A possible application of Theorem 3.28 is to add fast rejection criteria. A straightforward rejection filter is to check the number of bidirected edges, i.e., $n_2(G_B)$. However, computing $n_i(G_B)$ for moderate i can be expensive. It is worth noting, however, that a largest connected component can be found by a DFS/BFS traversal in $\mathcal{O}(p + |E_{\leftrightarrow}|)$ [Tarjan, 1972], which suggests checking $|D_{\max}| = |\tilde{D}_{\max}|$ as in Corollary 3.29. Another observation is that an algorithm to compute $n_i(G)$ is output sensitive, i.e., the time needed depends on the value of $n_i(G)$ itself. Komusiewicz and Sommer [2021] proposed an algorithm with solution delay $\mathcal{O}(i^2 \Delta)$ where Δ is the maximum degree in the input graph. Therefore, if G_B is sparse, one would expect a fast return of f .

3.6 Computational Algebra for Cyclic Graphs

We now revisit the problem of equivalence from a different perspective. Theorem 3.1 provides a tractable computational procedure for verifying equivalence with the aid of computer algebra systems, such as MACAULAY2 [Grayson and Stillman, 1993]. Notably, this approach does not require acyclicity assumptions and therefore extends naturally to cyclic graphs.

The main challenge lies in handling the quadratic terms in the equations: Lemma 3.2 implies that A_{ij} is an affine linear function of $\tilde{\lambda}_{\bullet,i}$, where $\tilde{\lambda}_{\bullet,i}$ denotes the collection $\{\lambda_{ji} : j \in V \setminus i\}$. Therefore, the product $A_{uw_1} \cdot A_{vw_2}$ is at most quadratic in $\tilde{\lambda}$. We see that a quadratic term involving a single variable, such as $\tilde{\lambda}_{ki}^2$, cannot appear in the product, since this would imply $u = v$, contradicting $(u, v) \in \mathcal{J}$. Nevertheless, quadratic terms involving two distinct variables do appear in the equations. The issue is that this may admit non-real solutions, whereas symbolic

polynomial solvers often operate over the algebraic closure (typically $\mathbb{C}$). However, in our framework, we are only interested in solvability over $\mathbb{R}$. At first glance, the mixed quadratic terms distinguish the complex and real solvability of the system $\mathcal{L}$. The following lemma shows that, in the unrestricted domain (not requiring $\det(I - \tilde{\Lambda}) \neq 0$), the two notions of solvability are in fact equivalent due to the specific structure of the system.

Lemma 3.30. *For any $\Lambda \in \mathbb{R}^{G_D}$, the system $\mathcal{L}$ as defined in (3.1) is solvable in $\tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}$ if and only if it is solvable in $\tilde{\Lambda} \in \mathbb{C}^{\tilde{G}_D}$.*

Proof. For the “if” direction, assume there exists a solution $\tilde{\Lambda}^* \in \mathbb{C}^{\tilde{G}_D}$. Using Lemma 3.2, we can write the system $\mathcal{L}$ as:

$$\mathcal{L} = \{f_1 g_1 = 0, f_2 g_2 = 0, \dots, f_m g_m = 0\}, \quad (3.83)$$

where $f_1, \dots, f_m, g_1, \dots, g_m$ are affine linear in $\tilde{\Lambda}$. We define $S \subseteq [m]$ to be the subset that $f_s(\tilde{\Lambda}^*) = 0$ for all $s \in S$. It follows that $g_r(\tilde{\Lambda}^*) = 0$ for all $r \in [m] \setminus S$. Since $\{f_s = 0 : s \in S\} \cup \{g_r = 0 : r \in [m] \setminus S\}$ is a linear system in $\tilde{\Lambda}$, the existence of a complex solution $\tilde{\Lambda}^* \in \mathbb{C}^{\tilde{G}_D}$ implies the existence of a real one. The other direction is clear. $\square$

Lemma 3.30 cannot be directly adapted to the restricted domain $\mathbb{R}_{\text{reg}}^{\tilde{G}_D}$ as checking $\det(I - \tilde{\Lambda}) \neq 0$ involves checking a nonlinear function. However, when considering only ADMGs, Lemma 3.30 implies that checking solvability of $\mathcal{L}$ over the complex is both necessary and sufficient for model inclusion, while cyclic DMGs require an extra determinant verification. We see that the system $\mathcal{L}$ is linear in the sense that it is solvable if and only if there is a corresponding solvable linear system $\{f_s = 0 : s \in S\} \cup \{g_r = 0 : r \in [m] \setminus S\}$. This fact is critical for equivalence testing: instead of checking generic solvability, we can randomly sample one $\Lambda^* \in \mathbb{R}^{G_D}$ and check solvability there once alone.

Proposition 3.31. *Let G and $\tilde{G}$ be two DMGs with the same vertex set V . Let $\Lambda^* \in \mathbb{R}^{G_D}$ be a random sample such that $\lambda^* \sim \text{Uniform}(-1, 1)$ for all nonzero entries. Let $\mathcal{L}(\Lambda^*)$ be the system after substituting $\Lambda = \Lambda^*$ into $\mathcal{L}$. Then*

$$\mathbb{P}\left(\mathcal{M}(G) \subseteq \overline{\mathcal{M}(\tilde{G})}\right) = \begin{cases} 1 & \text{if } \mathcal{L}(\Lambda^*) \text{ is solvable in } \tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}, \text{ and} \\ 0 & \text{if } \mathcal{L}(\Lambda^*) \text{ is not solvable in } \tilde{\Lambda} \in \mathbb{R}^{\tilde{G}_D}. \end{cases} \quad (3.84)$$

Proof. Let us start with the case that for a random sampled $\Lambda^* \in \mathbb{R}^{G_D}$, $\mathcal{L}(\Lambda^*)$ is not solvable. Then the probability that $\mathcal{L}$ is still generically solvable is the probability that the random sampled Λ^* exactly landed at the exceptional null set of $\mathbb{R}^{G_D}$, which happens with probability 0. Therefore, $\mathcal{M}(G) \not\subseteq \overline{\mathcal{M}(\tilde{G})}$ almost surely. Now assume that for a random sampled $\Lambda^* \in \mathbb{R}^{G_D}$, $\mathcal{L}(\Lambda^*)$ is solvable. Following the proof of Lemma 3.30, we know that there exists an $S \subseteq [m]$ so that $\mathcal{L}$ can be reduced to $\{f_s = 0 : s \in S\} \cup \{g_r = 0 : r \in [m] \setminus S\}$. For a linear system, solvability happens generically, meaning that such system is either solvable for almost all parameter choice or for almost no parameter choice. Assume that such linear system is actually generically unsolvable, then we would have landed again at the exceptional null set with the random sample, which happens with probability zero. Therefore, the corresponding linear system must be generically solvable almost surely, implying that $\mathcal{L}$ is indeed solvable for any generic $\Lambda \in \mathbb{R}^{G_D}$ almost surely. $\square$

The above observations lead to the following algorithm (Algorithm 2), which has the advantage of naturally accommodating cyclic graphs (code available here). It is worth noting that in order

Algorithm 2: Check if two DMGs G and $\tilde{G}$ are model equivalent using randomization.

Input: DMGs $G, \tilde{G}$ with directed edges encoded in $\Lambda, \tilde{\Lambda}$

Output: Whether $\overline{\mathcal{M}(G)} = \overline{\mathcal{M}(\tilde{G})}$

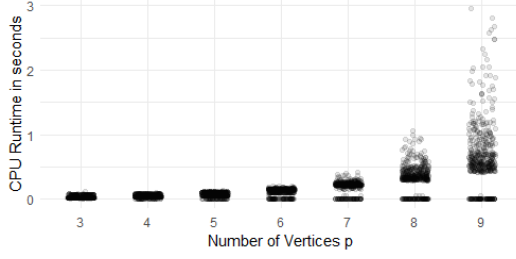
```

1 Procedure FastRejection( $G, \tilde{G}$ );
2   #Check numbers of bidirected edges and the size of maximal district
3   if  $|E_{\leftrightarrow}(G)| \neq |E_{\leftrightarrow}(\tilde{G})|$  or  $|D_{\max}(G)| \neq |D_{\max}(\tilde{G})|$  then
4     return false;
5   end
6   else
7     return true;
8   end

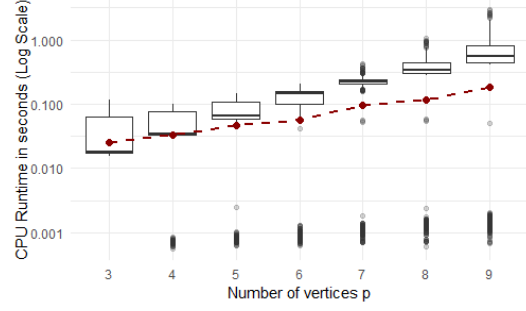
9 Procedure CheckInclusion( $G, \tilde{G}$ );
10  Add variable  $\tilde{\lambda}_{ij} \in \mathbb{R}$  foreach  $i \rightarrow j \in \tilde{G}_D$ . Add variable  $t \in \mathbb{R}$ ;
11  #Randomization
12  Substitute  $\Lambda \sim \text{Uniform}(-C, C)$  for some  $C > 0$ ;
13   $A \leftarrow (I - \tilde{\Lambda})^T (I - \Lambda)^{-T}$ ;
14  #Initialize the equation system  $L$ 
15   $L \leftarrow \{\emptyset\}$ ;
16  foreach absent bidirected edge  $u \leftrightarrow v$ ,  $u \neq v$  in  $\tilde{G}$  do
17    foreach bidirected edge  $w_1 \leftrightarrow w_2$  or  $w_1 = w_2$  in  $G$  do
18      if  $A_{uw_1} \neq 0$  and  $A_{vw_2} \neq 0$  then
19        add constraint  $A_{uw_1} A_{vw_2} = 0$  to  $L$ ;
20      end
21    end
22  end
23  #Ensure nonzero determinant
24  Add constraint  $t \cdot \det(I - \tilde{\Lambda}) - 1 = 0$  to  $L$ ;
25  #Determine inclusion by checking generic solvability
26  if there is a solution  $(\tilde{\Lambda}, t) \in \mathbb{R}^{\tilde{G}_D} \times \mathbb{R}$  to  $L$  then
27    return true;
28  end
29  else
30    return false;
31  end

32 Main Procedure CheckEquivalence( $G, \tilde{G}$ );
33  if FastRejection( $G, \tilde{G}$ ) = false then
34    return  $\overline{\mathcal{M}(G)} \neq \overline{\mathcal{M}(\tilde{G})}$ ;
35  end
36  #Check both directions
37  if CheckInclusion( $G, \tilde{G}$ ) and CheckInclusion( $\tilde{G}, G$ ) then
38    return  $\overline{\mathcal{M}(G)} = \overline{\mathcal{M}(\tilde{G})}$ ;
39  else
40    return  $\overline{\mathcal{M}(G)} \neq \overline{\mathcal{M}(\tilde{G})}$ ;
41  end

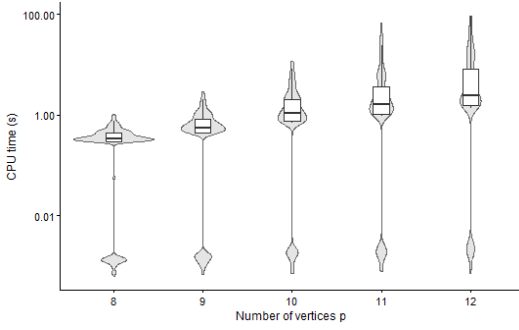
```



(a) Raw data of CPU time using Algorithm 2, for $p \in \{3, 4, \dots, 9\}$.



(b) CPU time in log scale using Algorithm 2. The Red dashed line represents the average CPU time.



(c) Distribution of the CPU time (with the y-axis in log scale) for moderate size of graphs, i.e., $p \in \{8, 9, 10, 11, 12\}$, using Algorithm 2.

$ E_{\leftrightarrow} $	Mean	Median	Min	Max
3	0.122	0.0473	0.000692	0.935
4	0.0890	0.00119	0.000752	0.540
5	0.0971	0.00135	0.000769	0.402
6	0.0950	0.00135	0.000874	0.444
7	0.114	0.00139	0.000975	0.350
8	0.156	0.222	0.000982	0.401
9	0.184	0.230	0.00109	0.360
10	0.216	0.232	0.00102	0.426
11	0.233	0.232	0.00106	0.482
12	0.241	0.235	0.00115	0.433
15	0.244	0.237	0.164	0.411
18	0.238	0.228	0.162	0.441

(d) Execution time in seconds with different numbers of bidirected edges, where we fix $|V| = 7$.

Figure 18: Performance analysis of Algorithm 2.

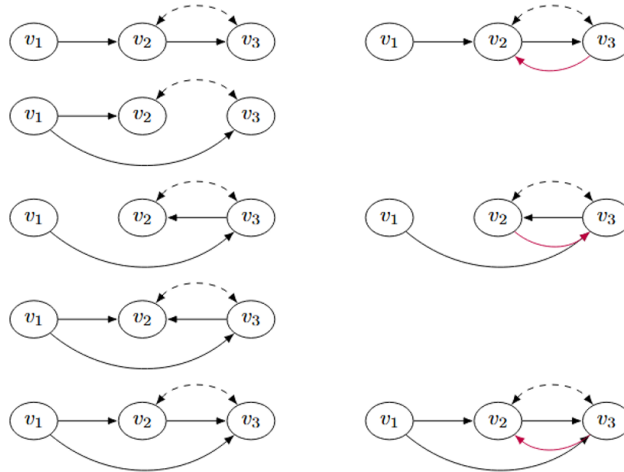


Figure 19: The equivalence class of the IV graph when cycles are allowed. Let the five ADMGs on the left hand side be $G_1, \dots, G_5$. Let the three cyclic DMGs on the right hand side be G_6, G_7, G_8 .

to control the determinant, we extended our equation system $\mathcal{L}_{cyc} = \mathcal{L} \cup \{t \cdot \det(I - \tilde{\Lambda}) - 1 = 0\}$ with extended variables $(\tilde{\Lambda}, t) \in \mathbb{R}^{\tilde{G}_D} \times \mathbb{R}$.

To evaluate the performance of Algorithm 2, we run random experiments for different numbers of vertices ($p = 3, 4, \dots, 9$) and record for each case the **CPU processing time** in seconds. For each fixed p , we repeat the process of randomly sampling two ADMGs with the same number of bidirected edges for 500 times. To show the result as raw as possible, we record the runtime as in Figure 18a, where each data point represents an experiment. Figure 18b shows the computation time on a log scale and demonstrates an exponential increase in average CPU time.

The distribution of CPU time is bimodal after adding rejection filters (Figure 18c). Figure 18d is obtained by fixing $p = |V| = 7$ and varying the number of bidirected edges. For each different $|E_{\leftrightarrow}|$, we sample 200 pairs of ADMGs with the given $|E_{\leftrightarrow}|$ and run Algorithm 2 on these pairs. We record the mean, median, minimum, and maximum CPU processing time of these samples. Since the size of the equation system $|L|$ in Algorithm 2 decreases as the number of bidirected edges increases, we observe a more extreme maximal CPU time when $|E_{\leftrightarrow}|$ is small. At the same time, when $|E_{\leftrightarrow}|$ is small, the fast rejection via maximal connected component is more likely to happen as we observe also a more extreme minimal CPU time.

To demonstrate an equivalent class separation, we separated all $25 \times 2^{\binom{3}{2}} = 200$ ADMGs with 3 vertices into 74 equivalence classes (see here). 25 classes among them consist of a unique ADMG without bidirected edges. These are the 25 DAGs with three vertices (which are identifiable for linear non-Gaussian models). Another trivial class is the one that consists of all 25 ADMGs with fully connected bidirected parts.

The motivation for using computer algebra methods such as Algorithm 2 is that such algorithms naturally accommodate cycles by additionally checking the determinant condition. We illustrate the standard IV equivalence class below and compare it with the equivalence class in the lv-LiNG case, such as the one considered in [Dai et al., 2026].

Example 3.9. Consider again the IV equivalence class. When cycles are allowed, the equivalence class (Figure 19) is extended by three additional DMGs containing a loop between v_2 and v_3 . Consider the equivalence class of the IV graph (G_1) in a related setting, such as lv-LiNG, where confounding effects are only allowed to enter the distribution linearly [Dai et al., 2026]. For simplicity, assume that there is only one latent variable L_1 with directed edges to v_2 and v_3 . Then, the latent projection of the resulting equivalence class consists only of G_1 and G_5 . We see that allowing nonlinear confounding leads to much richer structures, and our approach has the advantage of not requiring explicit relations between latent variables (see Figure 20).

We note that for the specific case in Example 3.9, the “traversal conjecture” holds as shown in Figure 21. This is however, not always true in the cyclic case:

Example 3.10. Let us revisit the counterexample to the traversal property in Figure 16. If we generalize it to the cyclic setting, the equivalence class is enriched by two additional DMGs, illustrated in Figure 22. It is straightforward to verify that there is no way to traverse the entire equivalence class by simply adding, removing or reversing edges, thereby disproving the traversal conjecture.

Even though we have shown that the transformation cannot be achieved using only edge deletion and addition, there might still be other ways of locally traversing the equivalence class. In that case, one could use BFS/DFS to enumerate the equivalent graphs. Similar results have been proved in related settings using tools from transversal matroid theory [Mason, 1972, Ingleton and Piff, 1973, Dai et al., 2026]. We leave this as a direction for future research.

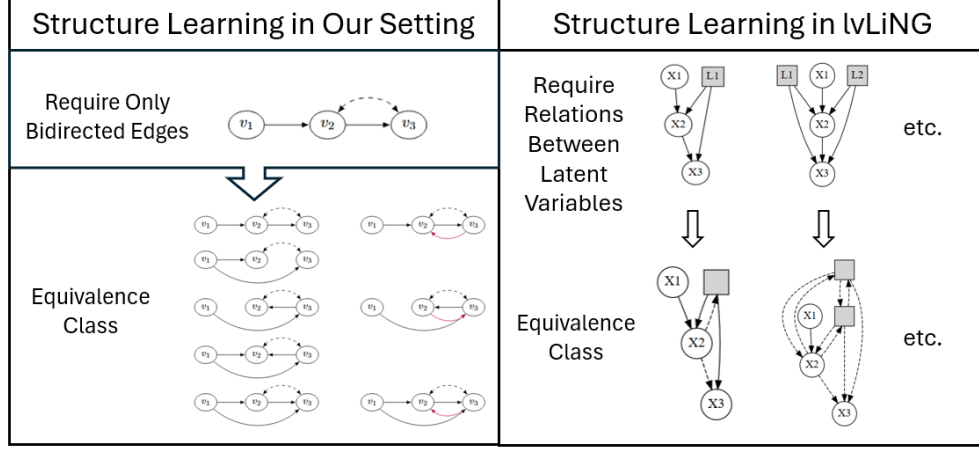


Figure 20: Comparison of structure learning in linear and nonlinear confounded settings. The equivalence classes in lvLiNG are represented by maximally inclusive digraphs [Dai et al., 2026], where solid edges are present in all digraphs, while dashed edges are present in some but not all digraphs.

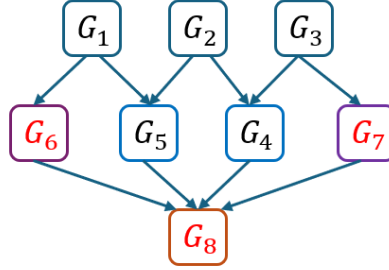


Figure 21: Hasse diagram of $G_1 \dots, G_8$ from Figure 19. A directed edge from one DMG to another represents that one could be derived from another by adding a directed edge.

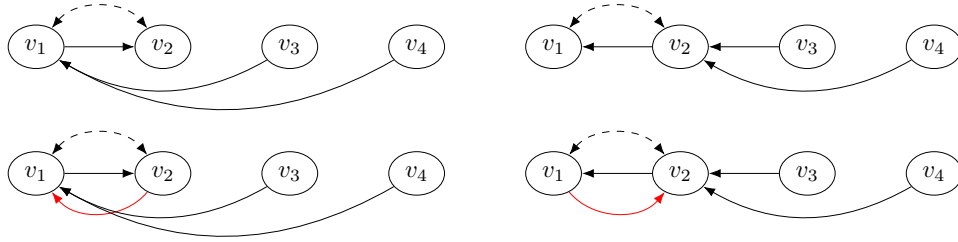


Figure 22: An example of a disconnected equivalence class. The DMGs on the left-hand side cannot be transformed into those on the right-hand side by adding, removing or reversing directed edges while remaining within the equivalence class.

3.7 Solving via Satisfiability Modulo Theories

Until now, we have approached the problem of characterizing equivalence from two different perspectives. Starting from the theoretical result of Theorem 3.1, we developed graphical characterizations and corresponding algorithms for our main task. On the other hand, we showed that the characterization in Theorem 3.1 can also be implemented directly using computer algebra software. While this approach has the advantage of naturally accommodating cycles, it comes at the cost of being computationally infeasible for larger graphs.

In this section, we formulate the inclusion problem as a satisfiability problem, more specifically as an SMT (Satisfiability Modulo Theories) problem [Biere et al., 2009, Barrett and Tinelli, 2018]. As we will see, our characterization naturally admits such a formulation. We will then show that this approach leads to the most practically efficient algorithm among the methods considered.

Recall that Theorem 3.1 characterizes model inclusion $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$ in terms of the solvability of an equation system of the following form:

$$\begin{cases} f_1 g_1 = 0 \\ f_2 g_2 = 0 \\ f_3 g_3 = 0 \\ \dots \\ f_k g_k = 0, \end{cases} \quad (3.85)$$

where all $f_1, \dots, f_k, g_1, \dots, g_k$ are linear. As argued in Lemma 3.3, the system is solvable if and only if for each $i \in [k]$ we either choose f_i or g_i so that the chosen linear system is solvable. This corresponds to determining the feasibility of the following logical form

$$(f_1 = 0 \vee g_1 = 0) \wedge (f_2 = 0 \vee g_2 = 0) \wedge \dots \wedge (f_k = 0 \vee g_k = 0). \quad (3.86)$$

This is an SMT problem, i.e., the problem of determining whether a mathematical formula is satisfiable. Note that SMT generalizes the Boolean satisfiability problem (SAT) to formulas involving richer theories, such as real arithmetic and various data structures. The above problem can be encoded as an SMT formula and then solved using an SMT solver such as Z3 [de Moura and Bjørner, 2008]. Based on this idea, we designed the SMT-based workflow shown in Algorithm 3 (code available here).

At a high level, the system typically contains far more equations than variables, as the number of equations grows on the order of $\mathcal{O}(|V|^4)$, whereas the number of variables grows on the order of $\mathcal{O}(|V|^2)$. For such an overdetermined system, an SMT solver incrementally constructs a Boolean assignment to the theory atoms and invokes a theory solver to check whether the corresponding linear constraints are consistent. If an inconsistency is detected, the solver derives a conflict clause, thereby ruling out the current assignment and many related assignments. Otherwise, the solver continues exploring refinements of the current partial assignment. Since the learned clauses prune large portions of the search space, the search tree is often dramatically reduced, and Z3 rarely needs to explore all possible assignments. Figure 23 demonstrates the practical performance of Algorithm 3, which uses Z3’s modern SAT branching heuristics (e.g., VSIDS [Moskewicz et al., 2001], LRB [Liang et al., 2016], and restarts [Audemard and Simon, 2012]) within its CDCL(T) (Conflict-Driven Clause Learning modulo Theories) architecture [Nieuwenhuis et al., 2006, Marques Silva and Sakallah, 1996, 1999, Bayardo and Schrag, 1997]. Note that after preprocessing, conflict learning is handled directly by Z3. Since the design and implementation of modern SMT solvers are well beyond the scope of this work, we do not

Algorithm 3: An SMT based approach

Input: Two ADMGs G and $\tilde{G}$
Output: Whether $\mathcal{M}(G) \subseteq \mathcal{M}(\tilde{G})$

```

1 #Preprocessing
2 Add  $x_{u,v} \in \mathbb{R}$  to var foreach  $u \rightarrow v \in \tilde{G}$ ;
3 Sample a random  $\Lambda \in \mathbb{R}^{G_D}$ ;
4 Compute  $B = (I - \Lambda)^{-1}$ ;
5 Compute  $\mathcal{J} \otimes \mathcal{K} \subseteq V^4$ ;
6  $i = 1$ ;
7 foreach  $(p, q, r, s) \in \mathcal{J} \otimes \mathcal{K}$  do
8   compute  $f_i = b_{pq} - \sum_{k \rightarrow p \in \tilde{G}} x_{kp} b_{kq}$ ;
9   compute  $g_i = b_{rs} - \sum_{k \rightarrow r \in \tilde{G}} x_{kr} b_{ks}$ ;
10   $i \leftarrow i + 1$ ;
11 #Determine satisfiability using SMT
12 Initialize: SOLVER();
13 foreach  $i$  do
14   pass if  $f_i = 0$  or  $g_i = 0$ ;
15   if  $f_i$  is nonzero constant then
16     if  $g_i$  is nonzero constant then
17       return unsatisfiable;
18     else
19       add constraint  $f_i = 0$  to SOLVER();
20   if  $g_i$  is nonzero constant then
21     add constraint  $f_i = 0$  to SOLVER();
22   else
23     add  $b_i \in \text{BOOLEAN}$  to var;
24     add constraint  $(b_i, f_i = 0, g_i = 0)$  to SOLVER();
25 Run SOLVER() with respect to var;

```

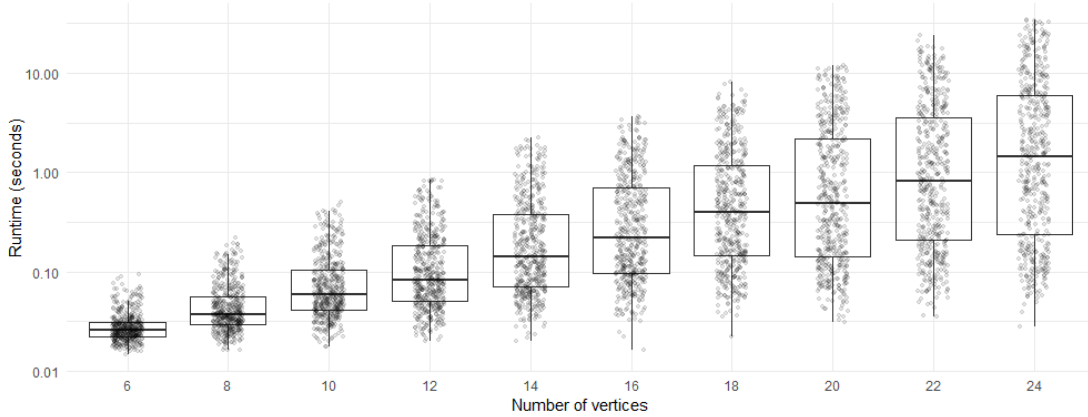


Figure 23: Performance of an SMT solver

attempt to describe the underlying algorithms employed by Z3. Instead, our interest lies in showing that our problem can be naturally encoded as an SMT instance and solved effectively using an off-the-shelf SMT solver.

To the end, we want to emphasize that an SMT-based approach does not necessarily guarantee an overall more favorable time complexity, as we distinguish between the following notions:

- **Worst-case complexity:** This is independent of any distributional assumptions.
- **Average-case complexity:** This is only meaningful once a distribution over the input instances has been specified.
- **Practical performance:** This measures the runtime on a given set of benchmark instances.

Note that average-case complexity is generally difficult to determine, as even uniformly sampling ADMGs is both technically challenging and unnecessary for evaluating the practical performance of our algorithms. Although the worst-case complexity remains exponential, clause learning, conflict analysis, and incremental linear feasibility checking can prune large portions of the search space in practice, resulting in substantially better empirical performance on many instances. Compared with Algorithm 2, which directly solves the underlying quadratic systems, the SMT-based approach demonstrates superior performance and is able to consistently handle much larger graphs, with up to around 24 vertices. The main limitation is that the SMT-based approach is not directly applicable to cyclic graphs. In the benchmark against Algorithm 1, the SMT-based approach also achieves better overall efficiency, as its architecture combined with the solver’s internal branching heuristics dynamically reduces the search space, whereas Algorithm 1 requires enumerating all minimal vertex covers.

4 Conclusion

In this thesis, we focus on the equivalence of linear non-Gaussian causal models in the presence of general latent confounding, where “general” means that the latent confounding is allowed to enter the distribution nonlinearly, subject only to a genericity condition (Equation (2.5)). This genericity condition allows us to characterize equivalent models by solving a system of quadratic equations (Theorem 3.1). Building on this result, our main contributions are the following:

1. A graphical criterion for equivalence (Theorem 3.7) and its dual version (Theorem 3.11). Moreover, we establish a practically efficient algorithm (Algorithm 1) and provide an implementation (source code).
2. Local characterizations of equivalence under directed edge removal (Theorem 3.23) and reversal (Theorem 3.24).
3. A study of the connectedness constraints on the confounding structure (Theorem 3.28).
4. A computer algebra method (Algorithm 2) for determining equivalence of cyclic graphs and its implementation (source code).
5. An SMT-based strategy (Algorithm 3) and its implementation (source code).

4.1 Limitations and Future Work

While our results provide an elegant graphical characterization of equivalence for non-Gaussian models under general latent confounding, verifying the characterization can be computationally demanding for large graphs. The main computational challenge arises from an enumeration step involving inclusive minimal vertex covers. Although the enumeration can be performed with polynomial delay, the number of inclusive minimal vertex covers can grow exponentially with the size of the graph. Consequently, the overall procedure may become computationally expensive in the worst case, even though in many instances only a single candidate needs to be identified. We argue that this limitation arising from the use of vertex covers is natural in view of Theorem 3.1, which shows that determining model inclusion in our setting is equivalent to checking the solvability of a system of quadratic equations. In contrast, graphical criteria are typically associated with rank constraints, which are exclusively linear. We therefore developed the vertex-cover-based selection criterion to effectively reduce the quadratic system to a linear one.

Although the results in Theorem 3.1 and Theorem 3.28 are valid for DMGs, the graphical results in Theorem 3.7, Theorem 3.23 and Theorem 3.24 do not always extend to the cyclic case. This limitation stems from the fact that allowing cycles requires us to consider the determinant of the directed edge matrix, which introduces nonlinear equations involving higher-degree polynomials. To avoid this complication, we have chosen to focus exclusively on ADMGs (see example 3.4).

We showed that the traversal conjecture is false in both the cyclic (Figure 22) and acyclic settings (Figure 16). Therefore, an interesting direction for future work is to develop transformation characterizations that incorporate other local graphical operations. Another open problem is to determine whether Conjecture 1 holds. A proof of Conjecture 1 would yield a simple characterization of edge reversal. Moreover, we studied the connectedness constraints on the confounding structure. These constraints are necessary for equivalence and are intuitive, but they can be computationally difficult to verify for large graphs. A natural direction for future work is therefore to understand how the bidirected part of a causal graph can vary within its equivalence class.

To conclude, this thesis addressed the fundamental question of when two causal graphs are model equivalent through graphical and algebraic results, as well as the development of computational methods for answering these questions, thereby providing a necessary step toward principled causal discovery methods in the presence of general latent confounding.

References

- Jeffrey Adams, Niels Hansen, and Kun Zhang. Identification of partially observed linear causal models: Graphical conditions for the non-Gaussian and heterogeneous cases. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 22822–22833. Curran Associates, Inc., 2021.
- Ayesha R Ali, Thomas S Richardson, Peter L Spirtes, and Jiji Zhang. Towards characterizing Markov equivalence classes for directed acyclic graphs with latent variables. *arXiv preprint arXiv:1207.1365*, 2005.
- Carlos Améndola, Mathias Drton, Alexandros Grosdos, Roser Homs, and Elina Robeva. Third-order moment varieties of linear Non-Gaussian graphical models. *Information and Inference: A Journal of the IMA*, 12(3):iaad007, 2023.
- Steen A. Andersson, David Madigan, and Michael D. Perlman. A characterization of Markov equivalence classes for acyclic digraphs. *The Annals of Statistics*, 25(2):505 – 541, 1997.
- Gilles Audemard and Laurent Simon. Refining restarts strategies for SAT and UNSAT. In Michela Milano, editor, *Principles and Practice of Constraint Programming*, pages 118–126, Berlin, Heidelberg, 2012. Springer Berlin Heidelberg. ISBN 978-3-642-33558-7.
- Clark Barrett and Cesare Tinelli. *Satisfiability Modulo Theories*, pages 305–343. Springer International Publishing, Cham, 2018. ISBN 978-3-319-10575-8.
- Roberto J. Bayardo and Robert C. Schrag. Using CSP look-back techniques to solve real-world SAT instances. In *Proceedings of the Fourteenth National Conference on Artificial Intelligence and Ninth Conference on Innovative Applications of Artificial Intelligence*, AAAI’97/IAAI’97, page 203–208. AAAI Press, 1997. ISBN 0262510952.
- A. Biere, A. Biere, M. Heule, H. van Maaren, and T. Walsh. *Handbook of Satisfiability: Volume 185 Frontiers in Artificial Intelligence and Applications*. IOS Press, NLD, 2009. ISBN 1586039296.
- Simon Bing, Jonas Wahl, and Jakob Runge. Structural causal bottleneck models, 2026.
- Ruichu Cai, Zhiyi Huang, Wei Chen, Zhifeng Hao, and Kun Zhang. Causal discovery with latent confounders based on higher-order cumulants. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 3380–3407. PMLR, 23–29 Jul 2023.
- Li Chen, Rasmus Kyng, Yang P. Liu, Richard Peng, Maximilian Probst Gutenberg, and Sushant Sachdeva. Maximum flow and minimum-cost flow in almost-linear time. In *2022 IEEE 63rd Annual Symposium on Foundations of Computer Science (FOCS)*, pages 612–623, 2022.
- Tom Claassen and Joris M Mooij. Establishing Markov equivalence in cyclic directed graphs. In *Uncertainty in Artificial Intelligence*, pages 433–442. PMLR, 2023.
- Haoyue Dai, Immanuel Albrecht, Peter Spirtes, and Kun Zhang. Distributional equivalence in linear non-Gaussian latent-variable cyclic causal models: Characterization and learning, 2026.
- G. Darmon. Analyse générale des liaisons stochastiques. Etude particulière de l’analyse factorielle linéaire. *Rev. Inst. Internat. Statist.*, 21:2–8, 1953. ISSN 0373-1138.

- Leonardo de Moura and Nikolaj Bjørner. Z3: An efficient SMT solver. In C. R. Ramakrishnan and Jakob Rehof, editors, *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340, Berlin, Heidelberg, 2008. Springer Berlin Heidelberg. ISBN 978-3-540-78800-3.
- Mathias Drton. Algebraic problems in structural equation modeling. In *The 50th anniversary of Gröbner bases*, volume 77, pages 35–87. Mathematical Society of Japan, 2018.
- Mathias Drton and Marloes H. Maathuis. Structure learning in graphical modeling. *Annual Review of Statistics and Its Application*, 4(Volume 4, 2017):365–393, 2017. ISSN 2326-831X.
- Mathias Drton and Thomas S. Richardson. Binary models for marginal independence. *J. R. Stat. Soc. Ser. B Stat. Methodol.*, 70(2):287–309, 2008. ISSN 1369-7412,1467-9868.
- Mathias Drton, Marina Garrote-López, Niko Nikov, Elina Robeva, and Y. Samuel Wang. Causal discovery for linear non-Gaussian models with disjoint cycles. In Silvia Chiappa and Sara Magliacane, editors, *Conference on Uncertainty in Artificial Intelligence, Rio Othon Palace, Rio de Janeiro, Brazil, 21-25 July 2025*, volume 286 of *Proceedings of Machine Learning Research*, pages 1064–1083. PMLR, 2025a.
- Mathias Drton, Benjamin Hollering, and Jun Wu. Identifiability of homoscedastic linear structural equation models using algebraic matroids. *Advances in Applied Mathematics*, 163:102794, 2025b. ISSN 0196-8858.
- Robin J. Evans. Margins of discrete Bayesian networks. *Ann. Statist.*, 46(6A):2623–2656, 2018. ISSN 0090-5364,2168-8966.
- Morten Frydenberg. The chain graph Markov property. *Scandinavian Journal of Statistics*, 17: 333–353, 1990.
- Ira Gessel and Gérard Viennot. Binomial determinants, paths, and hook length formulae. *Advances in Mathematics*, 58(3):300–321, 1985. ISSN 0001-8708.
- AmirEmad Ghassami, Alan Yang, Negar Kiyavash, and Kun Zhang. Characterizing distribution equivalence and structure learning for cyclic and acyclic directed graphs. In *International Conference on Machine Learning*, pages 3494–3504. PMLR, 2020.
- Clark Glymour, Kun Zhang, and Peter Spirtes. Review of causal discovery methods based on graphical models. *Frontiers in Genetics*, Volume 10 - 2019, 2019. ISSN 1664-8021.
- Chris Godsil and Gordon Royle. *Graphs*, pages 1–18. Springer New York, New York, NY, 2001. ISBN 978-1-4613-0163-9.
- Daniel R. Grayson and Michael E. Stillman. Macaulay2, a software system for research in algebraic geometry, 1993.
- Jürgen Herzog and Takayuki Hibi. *Monomial Ideals*. Springer London, London, 2011. ISBN 978-0-85729-106-6.
- Zhongyi Hu and Robin Evans. Faster algorithms for Markov equivalence. In *Conference on Uncertainty in Artificial Intelligence*, pages 739–748. PMLR, 2020.
- Aapo Hyvärinen, Kun Zhang, Shohei Shimizu, and Patrik O. Hoyer. Estimation of a structural vector autoregression model using non-Gaussianity. *J. Mach. Learn. Res.*, 11:1709–1731, 2010. ISSN 1532-4435.

- A.W Ingleton and M.J Piff. Gammoids and transversal matroids. *Journal of Combinatorial Theory, Series B*, 15(1):51–68, 1973. ISSN 0095-8956.
- David S. Johnson, Mihalis Yannakakis, and Christos H. Papadimitriou. On generating all maximal independent sets. *Information Processing Letters*, 27(3):119–123, 1988. ISSN 0020-0190.
- C. Jutten and P. Comon. *Handbook of Blind Source Separation*. Academic Press, Oxford, 2010. ISBN 978-0-12-374726-6.
- Yasuaki Kobayashi, Kazuhiro Kurita, Yasuko Matsui, and Hirotaka Ono. Enumerating minimal vertex covers and dominating sets with capacity and/or connectivity constraints. In Adele Anna Rescigno and Ugo Vaccaro, editors, *Combinatorial Algorithms*, pages 232–246, Cham, 2024. Springer Nature Switzerland. ISBN 978-3-031-63021-7.
- Christian Komusiewicz and Frank Sommer. Enumerating connected induced subgraphs: Improved delay and experimental comparison. *Discrete Applied Mathematics*, 303:262–282, 2021. ISSN 0166-218X.
- Steffan L. Lauritzen. *Graphical Models*. Oxford University Press, 1996.
- Jia Hui Liang, Vijay Ganesh, Pascal Poupart, and Krzysztof Czarnecki. Learning rate based branching heuristic for SAT solvers. In Nadia Creignou and Daniel Le Berre, editors, *Theory and Applications of Satisfiability Testing – SAT 2016*, pages 123–140, Cham, 2016. Springer International Publishing. ISBN 978-3-319-40970-2.
- Bernt Lindström. On the vector representations of induced matroids. *Bulletin of the London Mathematical Society*, 5(1):85–90, 1973.
- Yiheng Liu, Elina Robeva, and Huanqing Wang. Learning linear non-Gaussian graphical models with multidirected edges. *Journal of Causal Inference*, 9(1):250–263, 2021.
- Marloes Maathuis, Mathias Drton, Steffen Lauritzen, and Martin Wainwright, editors. *Handbook of graphical models*. Chapman & Hall/CRC Handbooks of Modern Statistical Methods. CRC Press, Boca Raton, FL, 2019. ISBN 978-1-4987-8862-5.
- Takashi Nicholas Maeda and Shohei Shimizu. RCD: Repetitive causal discovery of linear non-Gaussian acyclic models with latent confounders. In *International Conference on Artificial Intelligence and Statistics*, 2020.
- J.P. Marques Silva and K.A. Sakallah. GRASP-a new search algorithm for satisfiability. In *Proceedings of International Conference on Computer Aided Design*, pages 220–227, 1996.
- J.P. Marques Silva and K.A. Sakallah. GRASP: a search algorithm for propositional satisfiability. *IEEE Transactions on Computers*, 48(5):506–521, 1999.
- J. H. Mason. On a class of matroids arising from paths in graphs. *Proceedings of the London Mathematical Society*, s3-25(1):55–74, 1972.
- Christopher Meek. Graphical models: selecting causal and statistical models, 1997.
- Karl Menger. Zur allgemeinen Kurventheorie. *Fundamenta Mathematicae*, 10(1):96–115, 1927.
- Joris M Mooij and Tom Claassen. Constraint-based causal discovery using partial ancestral graphs in the presence of cycles. In *Conference on Uncertainty in Artificial Intelligence*, pages 1159–1168. Pmlr, 2020.

- John W. Moon and Leo Moser. On cliques in graphs. *Israel Journal of Mathematics*, 3:23–28, 1965.
- M.W. Moskewicz, C.F. Madigan, Y. Zhao, L. Zhang, and S. Malik. Chaff: engineering an efficient SAT solver. In *Proceedings of the 38th Design Automation Conference (IEEE Cat. No.01CH37232)*, pages 530–535, 2001.
- Robert Nieuwenhuis, Albert Oliveras, and Cesare Tinelli. Solving SAT and SAT modulo theories: From an abstract Davis–Putnam–Logemann–Loveland procedure to DPLL(T). *J. ACM*, 53(6): 937–977, November 2006. ISSN 0004-5411.
- Judea Pearl. Linear models: A useful “microscope” for causal analysis. *Journal of Causal Inference*, 1(1):155–170, 2013.
- Judea Pearl and Rina Dechter. Identifying independencies in causal graphs with feedback. *ArXiv*, 1996.
- Hazel Perfect. Applications of Menger’s graph theorem. *Journal of Mathematical Analysis and Applications*, 22(1):96–111, 1968. ISSN 0022-247X.
- Thomas Richardson. Markov properties for acyclic directed mixed graphs. *Scand. J. Statist.*, 30(1):145–157, 2003. ISSN 0303-6898,1467-9469.
- Thomas Richardson and Peter Spirtes. Ancestral graph Markov models. *The Annals of Statistics*, 30(4):962 – 1030, 2002.
- Thomas S. Richardson, Robin J. Evans, James M. Robins, and Ilya Shpitser. Nested Markov properties for acyclic directed mixed graphs. *Ann. Statist.*, 51(1):334–361, 2023. ISSN 0090-5364,2168-8966.
- Elina Robeva and Jean-Baptiste Seby. Multi-trek separation in linear structural equation models. *SIAM J. Appl. Algebra Geom.*, 5(2):278–303, 2021. ISSN 2470-6566.
- Dominik Rothenhäusler, Christina Heinze, Jonas Peters, and Nicolai Meinshausen. BACKSHIFT: Learning causal cyclic graphs from unknown shift interventions. *Advances in Neural Information Processing Systems*, 28, 2015.
- Alberto Roverato and Milan Studeny. A graphical representation of equivalence classes of AMP chain graphs. *Journal of Machine Learning Research*, 7(38):1045–1078, 2006.
- Saber Salehkaleybar, AmirEmad Ghassami, Negar Kiyavash, and Kun Zhang. Learning linear non-Gaussian causal models in the presence of latent variables. *Journal of Machine Learning Research*, 21(39):1–24, 2020.
- Daniela Schkoda, Elina Robeva, and Mathias Drton. Causal discovery of linear Non-Gaussian causal models with unobserved confounding. *arXiv preprint arXiv:2408.04907*, 2024.
- Shohei Shimizu. *Statistical causal discovery: LiNGAM approach*. SpringerBriefs in Statistics. Springer Japan, Tokyo, 2022. ISBN 978-4-431-55783-8. JSS Research Series in Statistics.
- Shohei Shimizu, Patrik O. Hoyer, Aapo Hyvärinen, and Antti J. Kerminen. A linear non-Gaussian acyclic model for causal discovery. *J. Mach. Learn. Res.*, 7:2003–2030, 2006.
- V. P. Skitovich. Linear forms of independent random variables and the normal distribution law. *Izv. Akad. Nauk SSSR Ser. Mat.*, 18:185–200, 1954. ISSN 0373-2436.

- Peter Spirtes. Building causal graphs from statistical data in the presence of latent variables. In *Logic, methodology and philosophy of science, IX (Uppsala, 1991)*, volume 134 of *Stud. Logic Found. Math.*, pages 813–829. North-Holland, Amsterdam, 1994. ISBN 0-444-89341-5.
- Peter Spirtes and Clark Glymour. An algorithm for fast recovery of sparse causal graphs. *Social Science Computer Review*, 9(1):62–72, 1991.
- Peter Spirtes and Thomas Richardson. A polynomial time algorithm for determining DAG equivalence in the presence of latent variables and selection bias. In *Proceedings of the 6th International Workshop on Artificial Intelligence and Statistics*, volume 12, 1996.
- Peter Spirtes and Thomas Verma. Equivalence of causal models with latent variables. *Carnegie Mellon University Tech Report*, 1992.
- Peter Spirtes, Clark Glymour, and Richard Scheines. *Causation, Prediction, and Search*. The MIT Press, 01 2001. ISBN 9780262284158.
- Seth Sullivant, Kelli Talaska, and Jan Draisma. Trek separation for Gaussian graphical models. *Ann. Statist.*, 38(3):1665–1685, 2010. ISSN 0090-5364,2168-8966.
- Kelli Talaska. Determinants of weighted path matrices. *arXiv: Combinatorics*, 2012.
- Robert Tarjan. Depth-first search and linear graph algorithms. *SIAM Journal on Computing*, 1(2):146–160, 1972.
- Jin Tian and Judea Pearl. Causal discovery from changes. *arXiv preprint arXiv:1301.2312*, 2001.
- Daniele Tramontano, Yaroslav Kivva, Saber Salehkaleybar, Mathias Drton, and Negar Kiyavash. Causal effect identification in LiNGAM models with latent confounders. In Ruslan Salakhutdinov, Zico Kolter, Katherine A. Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, volume 235 of *Proceedings of Machine Learning Research*, pages 48468–48493. PMLR / OpenReview.net, 2024.
- Daniele Tramontano, Yaroslav Kivva, Saber Salehkaleybar, Negar Kiyavash, and Mathias Drton. Causal effect identification in lvLiNGAM from higher-order cumulants. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*, volume 267 of *Proceedings of Machine Learning Research*. PMLR / OpenReview.net, 2025.
- Daniele Tramontano, Mathias Drton, and Jalal Etesami. Parameter identification in linear non-Gaussian causal models under general confounding. *The Annals of Statistics*, 54(2):957 – 981, 2026.
- Shuji Tsukiyama, Mikio Ide, Hiromu Ariyoshi, and Isao Shirakawa. A new algorithm for generating all the maximal independent sets. *SIAM Journal on Computing*, 6(3):505–517, 1977.
- Thijs van Ommen. Efficiently deciding algebraic equivalence of bow-free acyclic path diagrams. In Negar Kiyavash and Joris M. Mooij, editors, *Uncertainty in Artificial Intelligence, 15-19 July 2024, Universitat Pompeu Fabra, Barcelona, Spain*, volume 244 of *Proceedings of Machine Learning Research*, pages 3410–3424. PMLR, 2024.

- Thomas Verma and Judea Pearl. Equivalence and synthesis of causal models. In Piero P. Bonissone, Max Henrion, Laveen N. Kanal, and John F. Lemmer, editors, *UAI '90: Proceedings of the Sixth Annual Conference on Uncertainty in Artificial Intelligence, MIT, Cambridge, MA, USA, July 27-29, 1990*, pages 255–270. Elsevier, 1990.
- Y. Samuel Wang and Mathias Drton. Causal discovery with unobserved confounding and non-Gaussian data. *J. Mach. Learn. Res.*, 24:271:1–271:61, 2023.
- Yiqin Yang, Mohamed S Nafea, Negar Kiyavash, Kun Zhang, and AmirEmad Ghassami. Causal discovery in linear models with unobserved variables and measurement error. In *NeurIPS 2024 Causal Representation Learning Workshop*, 2024.
- Jiji Zhang. On the completeness of orientation rules for causal discovery in the presence of latent confounders and selection bias. *Artificial Intelligence*, 172(16):1873–1896, 2008. ISSN 0004-3702.

A Code Repository

The complete implementation accompanying this thesis is available in the following two GitHub repositories:

1. <https://github.com/YanchuZhang/GC-LiNGAM>
2. https://github.com/YanchuZhang/Model_Equivalence_cyclic

The first repository contains the implementation of the graphical algorithms developed in this thesis (Algorithm 1), as well as the source code for the SMT-based approach (Algorithm 3). The second repository contains the implementation of the computer algebra method for cyclic graphs (Algorithm 2) discussed in Section 3.6. The two methods are provided in separate repositories because the first repository is based on **Python**, while the computer algebra method is implemented in **Macaulay2**.

Instructions for using the developed tools and reproducing the experiments are provided in the respective repositories.